

Benchmarking Requirements for Trustworthy Adaptive Neurostimulation Algorithms: A Computational Methods Study in Which Tuned Classical Control Defeats Deep Fitted-Q

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Abstract

Adaptive neurostimulation papers typically claim that a learned controller improves on simple baselines. This methods study asks a prior question: what benchmarking requirements must such a claim satisfy? Working in a fully specified synthetic beta-burst control surrogate, we train a two-hidden-layer fitted-Q network and then stress the claim against the controls the literature often omits. Three requirements emerge, each backed by an executed experiment. R1 - tuned classical control: a PID controller with gains selected on disjoint tuning seeds achieves mean return -9.035 ± 1.119 against the neural controller's -10.440 ± 1.268 on 120 shared held-out seeds (paired bootstrap 95% CI $[-1.456, -1.354]$), and a model-predictive controller reaches -9.614 ; the learned controller clears always-off, fixed-high and beta-threshold baselines but fails the classical bar. R2 - model-uncertainty spanning: transferring the frozen neural controller to six parameter-perturbed simulator replicas shows a small sign-variable transfer gap (-0.28 to $+0.88$), yet fixed-gain PID remains ahead on every replica, so a claim must state which model family its uncertainty study spans. R3 - ablation identity: removing the energy penalty collapses the neural-PID gap to $+0.004$, showing the objective's difficulty lives in stimulation economics, while drift-coupled efficacy is a tested model coupling associated with a larger neural-PID gap (gap -1.07 coupled, -0.16 decoupled). In the synthetic surrogate, stimulation probability rises with its assigned beta state. This is consistency with an assumed feedback variable, not physiological validation. The learned controller still loses to tuned classical controls. Separately, 438 GEO whole-blood PD/control samples give weak discriminative signal (AUC 0.596) and no stimulation outcomes, and 55 public fixed-condition tremor traces contain no adaptive policy data: the data-readiness requirement is documented by demonstration. Native 512-neuron REINFORCE pilots collapse to constant actions (five ON and one OFF across six seed pairs), and a periodic half-duty open-loop control is never on the pooled four-policy reward envelope. These negative results retain their short-horizon and overlapping-training-seed limitations; none reproduces the published SAC protocol. The contribution is the requirements framework itself, instantiated end to end: no clinical claim is made anywhere in this report.

1 Treatment context and boundary

Nonclinical study. No result in this manuscript supports patient treatment decisions. Parkinson’s disease adaptive deep-brain stimulation (aDBS) adjusts electrical input to neurophysiological signals. A serious benchmark is DBS-Gym, which models Kuramoto neural ensembles with beta-band, spatial and drift features: <https://github.com/NevVerVer/DBS-Gym>. The initial surrogate did not install or run DBS-Gym; subsequent sections report limited native full-neuron control runs and an upstream-protocol PSD cross-check. Later native REINFORCE pilots use the 512-neuron spatial scale, but not the published SAC algorithm, episode protocol or evaluation replication. They show collapse rather than an effective learned controller. The surrogate isolates a tractable prototype controller with a different state, action, reward and timestep to expose engineering and safety questions, not to substitute a benchmark. No other MEGA27 13a disease is copied.

2 Data provenance

Parkinson blood-expression GEO GSE99039, ncbi.nlm.nih.gov/geo accession GSE99039 (full URL in `data/parkinson/provenance.json`), has 558 matrix columns. All 438 matrix columns with idiopathic Parkinson or control labels and first 48 finite measured probes yield 205 IPD and 233 controls. The source gzip is 63827685 bytes and SHA256, measured sample IDs and derived SHA256 are saved in `data/parkinson/provenance.json`. GSM accessions are independent identifiers but cannot be represented as 438 independent aDBS experiments. The 438 rows are used in a distinct descriptive probe and never passed to the simulated controller, since blood expression is not instantaneous beta neural feedback.

3 Formal surrogate

Let $s_t = (b_t, d_t, \delta_t)$ denote scaled beta amplitude, endogenous drive and electrode drift. Let the action index $a_t \in \{0, 1, 2\}$ map to $u_t \in \{0, .5, 1\}$. These equations exactly describe *our* assumed dynamics, not a fitted patient mechanism:

$$u_t = \{0, .5, 1\}_{a_t}. \quad (1)$$

$$\epsilon_t \sim \mathcal{N}(0, 0.05^2). \quad (2)$$

$$b_{t+1} = \text{clip}_{[0, 1.5]}(.85b_t + .12d_t + .04\delta_t - .19u_t(1 - .25\delta_t) + \epsilon_t). \quad (3)$$

$$d_{t+1} = \text{clip}_{[.05, 1]}(.94d_t + .045 + .035 \sin(8b_{t+1})). \quad (4)$$

$$\delta_{t+1} = \text{clip}_{[0, 1]}(.99\delta_t + .01 + .001 \sin(3b_{t+1})). \quad (5)$$

$$r_t = -b_{t+1}^2 - .12u_t^2 - .035\mathbf{1}(u_t > 0). \quad (6)$$

$$G = \sum_{t=0}^{69} r_t. \quad (7)$$

$$Q(s, a) = \mathbb{E}[r_t + .95 \max_{a'} Q(s_{t+1}, a') \mid s_t = s, a_t = a]. \quad (8)$$

$$f_\theta(s) = W_3 \text{ReLU}(W_2 \text{ReLU}(W_1 s + b_1) + b_2) + b_3. \quad (9)$$

$$Y_{i,a_i} = r_i + .95 \max_{a'} f_{\theta_{k-1}}(s'_i)_{a'}, \quad Y_{i,a \neq a_i} = f_{\theta_{k-1}}(s_i)_a. \quad (10)$$

$$\theta_k = \arg \min_{\theta} \sum_{i,a} \ell_{\text{Huber}}(f_\theta(s_i)_a - Y_{i,a}). \quad (11)$$

$$\hat{\Delta} = \frac{1}{120} \sum_{i=1}^{120} (G_i^{\text{neural}} - G_i^{\text{threshold}}). \quad (12)$$

$$\text{CI}_{.95} = \text{quantile}_{.025,.975}(\hat{\Delta}_1^*, \dots, \hat{\Delta}_{3000}^*). \quad (13)$$

$$\text{burst steps} = \sum_{t=0}^{69} \mathbf{1}(b_t > .5). \quad (14)$$

No adverse effects, stimulation contact location, sensory symptoms, hardware constraints or individualized therapy models appear in these equations. That omission makes the policy unfit for clinical interpretation.

4 Simulation protocol

Training uses 3000 independently sampled state/action/transition tuples from the custom process with seed 101, 12 successive fitted-Q target rounds of 20 epochs each, and a two-hidden-layer width-32 network. The 120 test episodes use seeds 60100 through 60219, all policies sharing per-seed initial state and noise; this paired setup reduces simulation variance but does not create independent patient validation. Horizon is 70. Comparators are always-off, always-high, and high amplitude iff beta exceeds 0.5. The fixed threshold and fixed high are intentionally simple controls, not the optimized DBS-Gym PID, PPO or SAC methods. The distinction blocks a false “benchmark victory” reading.

5 Simulated outcomes

Controller	Mean G	Mean burst steps	Mean energy
Neural fitted-Q	−10.440	0.367	34.204
Always off	−19.369	35.667	0
Fixed high	−11.226	0.367	70.000
Beta threshold	−15.365	11.967	12.542

The neural-versus-threshold paired mean difference is +4.925 with naive paired-bootstrap interval +4.825 to +5.021. That interval quantifies only test-seed noise given a *fixed chosen synthetic simulator*; it omits scientific model uncertainty, design choices and human variability. A larger reward can hide more energy or real-world harms. The neural controller uses nearly three times the

threshold energy, an honest trade-off. Fixed high suppresses model bursts equally well at double the neural energy, but no safety limit is modeled.

6 Descriptive molecular arm

48 measured GEO probes were extracted for each of 438 distinct GSM samples. A same-series 306/132 split with training-only centering gives held-out AUC 0.596 for IPD versus control. This is not a meaningful clinical biomarker validation; series batch, subject and cohort leakage risks persist. No molecular signal enters the controller, no treatment response is predicted from it and no causal bridge from peripheral blood to beta LFP has been demonstrated. Results and sample IDs are recorded in `results/parkinson_geo.json`.

7 Published comparison status and next pivot

DBS-Gym’s source describes three settings (spectral features, spatial partial observability and non-stationary drift), tests RL and standard controls and reports relative robustness. Our surrogate cannot reproduce its architecture or numeric metrics. Later native controls approach its physical scale, but do not include a comparable trained RL controller or six-run protocol. The benchmark gate is therefore explicitly *missing*. The within-project pivot is to run identical policies and seeds under the actual DBS-Gym environment, audit action and feature compatibility, add neural safety constraints and independent parameter sweeps, and obtain properly intervention-linked data before any translational claims. The earlier eight-tool tally was a smoke-test checkpoint, not a current final count; later data products are described individually. A fresh deduplicated used-tool audit and the substantive body-page gate remain open. PDF total pages do not establish either gate.

8 Within-project reward-cost pivot

An energy cost of 0.12 was chosen in the original surrogate. This assumption strongly affects what “better” means. We trained a second neural controller with energy coefficient 0.40 on an independently seeded 2,400-transition synthetic set, then compared it to a matched coefficient-0.12 controller on the same 100 new noise seeds 70200–70299. Evaluated with the original 0.12 reward, the 0.12 controller gives -9.761 with mean energy 37.635 and 0.5 beta-burst steps; the energy-conservative 0.40 controller gives -13.546 with energy 14.830 and 10.01 burst steps. Under the 0.40 evaluation reward, the former yields -20.299, the latter -17.698. The paired heavy-minus-original differences switch sign, from -3.785 to +2.600. This is sensitivity to a user-defined simulator penalty and a sharp tradeoff, not evidence for either therapy or an optimized medical utility. Full outputs: `src/parkinson_pivot.py` and `results/parkinson_pivot.json`.

9 Native DBS-Gym smoke test: configuration-induced negative

A limited smoke run executed the upstream SpatialKuramoto environment rather than merely reading its source. To constrain CPU cost, the setup reduced the published 512-neuron grid to eight neurons on a $2 \times 2 \times 2$ grid and four time steps in env0, with the upstream beta-band-power-plus-action reward. Fixed-off obtained total reward 0.0 and fixed-high -0.2 over four steps. This tiny grid and short observation window apparently provide no measurable beta-band reward signal; high action pays only its amplitude penalty. This is a negative of the reduced configuration, not evidence that aDBS is harmful or that fixed-off is optimal. Our prior synthetic neural controller was not transferred because the upstream observation and continuous action spaces differ. Native code, exact settings, Git snapshot and reward traces: `src/parkinson_native.py` and `results/parkinson_native_smoke.json`. Executed science components JAX, Diffrax, Gymnasium and DBS-Gym bring this project’s strict count to eight; they do not fulfill the published benchmark gate. A faithful 512-neuron, original horizon and matched-policies experiment remains required.

10 Full 512-neuron Env0 pilot toward published comparison

The published KDD Table 1 reports Env0 beta power as a percentage of off-stimulation, with off 100% (SD 27), fixed-high 19.8% (SD 1.9), and SAC 27.4% (SD 5.9) with 87.8% stimulation energy. Source: <https://arxiv.org/html/2505.09624>. These are external literature values, not ours. The repository env0 config uses 512 neurons, an eight-by-eight-by-eight spatial grid, and a 5000-unit, approximately 5555-step episode. The paper’s Table 1 evaluation instead uses 10 episodes of 1500 steps per Env0 evaluation and aggregates six evaluations; therefore a 5555-step single trajectory is not its published horizon. We executed exactly this neuron/grid scale with a 1000-step pilot on one frequency seed 222 for OFF (normalized action 0, physical 0 V) and high (+1, physical +5 V). The first attempted “off” run mistakenly supplied normalized -1 and in fact delivered -5 V. We corrected the label, retained this run as an explicit negative-five-volts pilot and reran true zero action rather than silently deleting it.

On these 1000-step pilots, the mean per-step low-beta reward component (coefficient 10000 times the upstream windowed beta power) was 19.9901 OFF, 6.7811 for +5 V and 8.0725 for -5 V. The +5 V / OFF ratio is 33.9%, versus the published multi-run 19.8%, but the measures are not guaranteed identical and our 1000-step one-seed ratio is not a benchmark reproduction. Per-episode global LFP beta-power ratios computed from concatenated 1000 action windows are 29.5% for +5 V / OFF; this differs from the upstream windowed reward metric and from the published normalization method. Both amplitudes use 5000 arbitrary physical-amplitude-step units, OFF zero. This pilot establishes native full-neuron oscillation and an action effect, but not the requested published-comparable evaluation: published multi-episode evaluation duration and initialization, multiple matched seeds, and published feature/power aggregation need completion. Code: `src/parkinson_env0_pilot.py`; exact outputs in `results/parkinson_env0_*.json`.

11 Published description versus repository evaluation settings

The published methods say Env0 assessment spans ten 1500-step episodes with changing initialization and Table 1 aggregates six evaluations. The checked upstream repository version does not transparently implement that exact description through a single Env0 config: its `environment/env_configs/env0.py` training dictionary sets `total_episode_len` to 5000 model units, while its `eval0` dictionary sets 1000 units. Given action duration 0.9 model units, these imply approximately 5555 and 1111 action steps respectively, not 1500. Moreover, the main block in `aDBS_RL/evaluate_HF_DBS.py` imports `eval_envs.list` from `env2` in the checked snapshot. The script does implement the filtered spectral aggregation reproduced below, but this import and duration mismatch prevent us from treating a naive execution as a faithful Env0 Table 1 evaluation. We do not know whether the authors used a different revision or unpublished launch configuration for the paper. We preserve the discrepancy rather than patching a guessed episode length and calling it a replication. Our one-seed 5555-step experiment is still a valid trace of the stated upstream training dictionary, but the scientific result it supports is correspondingly narrower.

12 One full-horizon native Env0 paired-control run

A reproducible checkpointed execution reached the upstream 5000-unit episode horizon (5555 action steps) at the actual 512-neuron Env0 grid, with matched seed 222, zero (OFF) versus normalized +1 (+5 V, high stimulation). The checkpoint procedure was verified first: a 50+50-step resumed run had exact zero difference in reward and sampled LFP from an uninterrupted 100-step run using the same seed. The full-horizon OFF mean per-step upstream beta reward component was 20.2143, while high stimulation was 4.5436, a ratio of 22.48%. The separate whole-episode concatenated LFP spectral power ratio was 20.97%. These are distinct estimands: the former is the mean of per-step upstream reward components, the latter an FFT of concatenated LFP samples. The published Table 1 Env0 high/zero beta ratio is 19.8% with standard deviation 1.9 across six evaluations. Our one-seed ratio is directionally similar but its uninterrupted 5555-step training-config trajectory does not match the published 10-by-1500-step evaluation structure or reproduce the multi-run mean and uncertainty. The run did not evaluate SAC, the target deep-RL controller, on upstream observations. Matching the benchmark requires multiple published evaluation configurations and an upstream-trained RL policy; the gate stays open. The full result JSONs preserve action conversion and sampled-LFP hashes; checkpoint code is `src/parkinson_env0_chunk.py` and the exact resume test is `src/parkinson_chunk_verify.py`.

13 Native observable-threshold pilot

The initial custom surrogate used a scalar beta state unavailable in the actual DBS-Gym observation. To begin replacing that invented state with live simulator evidence, a fixed threshold on the preceding upstream LFP window’s low-beta spectral power was run in 512-neuron Env0 for 1200 steps on seed 223. The heuristic threshold was 0.001 in the upstream signal’s units; this

number was chosen without independent calibration. The policy delivered high action in 65.8% of steps, expended 3950 amplitude-step units versus 6000 for always-high at that horizon, and attained mean upstream beta reward component 11.178. No matched-seed comparator was run for that seed and horizon, so no comparative benefit is claimed. This is a simple feedback baseline, not deep RL. The experiment exposes the need to separate beta-suppression, energy burden and seed effects before any neural training claim. Script and output: `src/parkinson_env0_adaptive.py`, `results/parkinson_env0_threshold_1200_seed223.json`.

14 Cohort split reveals molecular fragility

The same GSE99039 matrix provides its own TRAINING and VALIDATION cohort labels. We used all 293 labelled training rows (140 IPD, 153 controls) and 75 labelled validation rows (35 IPD, 40 controls), with train-only standardization of 48 measured probes. The cohort-heldout AUC was 0.566, lower than the full eligible-matrix random split AUC 0.596. The initial first-150 subset cohort AUC had been 0.421, an inverse-like negative, but it did not persist after including every eligible sample and cannot be generalized. We preserve that limited subset result in Git history; selection effects, possible batch structure and lack of an independent series prevent a stable biological claim. It also cannot validate adaptive DBS policies, which consume neural LFP traces rather than blood expression. Exact held-out accession IDs and counts: `results/parkinson_geo_cohort.json`.

15 Upstream filtered power aggregation changes the native comparison

The original upstream DBS-Gym evaluation script, `aDBS_RL/evaluate_HF_DBS.py`, does not report the raw power of a concatenated LFP or mean windowed reward component directly. Its `calc_psd_for_simple_eval` first applies a fifth-order Butterworth filter between 12 and 30 Hz to the full concatenated LFP (zero-phase filtering), calculates a one-sided real FFT power with its own factor-of-two normalization, smooths the frequency-domain array with a length-12 all-ones numerator filter and denominator five, and sums bins strictly inside 12.5–21 Hz at sample spacing 0.0005 s. Rather than rerun 11,110 native action steps, we reused the two 98,117-point matched-seed LFP checkpoint arrays and applied this stated upstream evaluation aggregation. A direct check using the upstream `band_pass_envelope` utility matched our copied filter exactly (zero numerical difference on OFF). The filter changes the values to OFF 0.00644353, +5 V 0.00177429, hence +5 V/OFF 27.536%. The ratio is considerably higher than our same-seed mean windowed reward ratio 22.48%, the raw full-LFP ratio 20.97%, and the paper’s six-evaluation mean 19.8% (SD 1.9). Those three local ratios have distinct definitions and none may be quietly substituted for another. The authors’ paper comparison runs 10 evaluation episodes of 1500 steps under varying initialization per evaluation (six Env0 evaluations), rather than one 5555-step trajectory; it also samples different evaluation settings, so this one matched seed cannot diagnose the source of the remaining difference. More importantly, there is still no native upstream-trained SAC or other RL controller. The explicit negative tightens the benchmark gap rather than closing it. The cached-

wave hashes, exact filter and numbers are preserved in `src/parkinson_published_psd_audit.py` and `results/parkinson_published_psd_seed222.json`.

16 Independent blood-expression dataset transfer: a negative result

GSE6613 is an independent older whole-blood cohort with 105 GSM sample columns, of which 50 are Parkinson disease and 22 healthy controls; 33 neurological disease controls are intentionally excluded from this binary task. The Affymetrix platform shares only 13 of the 48 arbitrary probes previously taken from GSE99039. We trained a logistic model from all 438 labeled GSE99039 samples using only these 13 shared probes, log2-transformed positive GSE6613 intensities without looking at target labels, and standardized with GSE99039 source statistics alone. AUC on the 72 independent Parkinson/healthy GSE6613 accessions was 0.534, close to chance, in contrast to the same-series 0.596 AUC on 48 probes and 0.566 within-GSE99039 cohort split. The feature set differs and the studies use distinct platforms, eras and case mixes; those three AUCs cannot be interpreted as a clean nested validation sequence. The weak independent result underscores failure to establish a stable blood diagnostic marker with the convenience feature choice, much less an adaptive stimulation marker. No LFP, stimulation actions, or patient response appears in GSE6613. Data source: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE6613>. Source gzip SHA256, all evaluated GSM accessions, 13 shared probe IDs and calculations: `results/geo_independent.json`, script `src/geo_independent.py`. The series extends disease-context data, not policy trajectory coverage.

17 Short native deep-RL pilot: low training dose and no benchmark claim

The earlier adaptive LFP controller was a threshold heuristic rather than deep RL. We therefore implemented a small two-hidden-layer tanh policy, four observable inputs from the preceding actual upstream LFP window (beta spectral power, amplitude dispersion, window mean, previous action), and a Bernoulli choice between true OFF and +5 V. Four independent 180-step, full-512-neuron Env0 training episodes used reward-to-go REINFORCE with 0.99 discount, within-episode advantage centering, a 0.01 learning rate and gradient clipping; these choices were pilot engineering decisions, not published SAC hyperparameters. Checkpointing weights after training avoided a timeout rerunning the 720 expensive action steps before evaluation. Their training seed mean per-step rewards were -9.678, -7.746, -13.874, -5.645 with mixed stimulation rates between 0.417 and 0.544. These numbers across different seeds do not show learning: there are only four episodes, and the apparent last-episode gain can arise from the seed. In a separate fixed, deterministic 1000-step evaluation on seed 222, the resulting policy’s mean per-step reward was -13.554, mean beta component 13.529 and action-on fraction 0.500 with 2500 amplitude-step energy units. Its return is not compared to the preexisting 1000-step fixed-control runs as a formal policy gain because the training protocol, device-energy convention, and outcome measures need careful matched rechecks; even

a direct comparison would not create six-run uncertainty. We checked learned parameters against initialization: the largest L2 drift was just 0.000029 in output bias and weight-layer L2 shifts were below 0.000001. Decision-probe probabilities at representative inputs remain around 0.500 and depend heavily on previous action. This fails to establish meaningful learning; the deterministic 0.500 action-on rate reflects threshold artifacts more readily than useful LFP response. The REINFORCE update and neural policy did run in the native 512-neuron environment, but this is not an effective learned controller, *not* published six-evaluation, 10-by-1500-step SAC, and no clinical or policy-trajectory validation. Code: `src/parkinson_native_reinforce.py`; exact metrics, seeds and weight checksum: `results/parkinson_native_reinforce.json`.

18 Within-project optimizer pivot and a second native negative

The first native REINFORCE pilot barely changed its network weights. A bounded pivot removed the prior-action feature (fixing its input to zero), raised network weight initialization scale from 0.10 to 0.35, and used bias-corrected Adam on the episode policy gradient with step size 0.03. Three new 220-step native 512-neuron Env0 training episodes ran on seeds 820–822; the mean rewards -12.110, -7.995 and -14.474 again are different seeds, not a learning curve. Weight L2 movement became visible, 0.293 for the first layer and 0.390 for the second layer. But deterministic 1000-step evaluation at the matched seed 222 chose OFF at every step, yielding mean reward -19.990, exactly matching the archived 1000-step always-off control; fixed +5 V had -6.831 on this same seed/horizon. The controller collapsed to an inferior fixed policy. This single-seed result does not show a robust medical tradeoff. This pivot resolves the tiny-gradient implementation symptom but *does not* meet the learned-control benchmark. Code, trained weights and exact outcome are in `src/parkinson_native_reinforce_adam.py`, `data/parkinson/native_reinforce_adam_weights.npz` and `results/parkinson_native_reinforce_adam.json`. Longer, multi-seed work with a continuous-action method is needed; the six-run SAC benchmark remains unreplicated.

19 Public fixed-condition Parkinsonian tremor traces, not adaptive policy data

PhysioNet hosts human Parkinsonian finger-tremor velocity recordings from 16 people implanted with DBS, under effective DBS on/off and dopaminergic medication on/off conditions: <https://physionet.org/content/tremordb/1.0.0/>. We fetched the 55 original plain-text recordings in these eight resting-condition categories and numerically analyzed 15 represented subjects. Header file `rofh.he` lists `v5rof.rit`, while the actual directory and source description list `v5rof.let`; the script explicitly corrects that one source-header discrepancy instead of silently losing a trace. Welch power in the 4–6 Hz tremor band was calculated on each 60-second, 100-Hz raw instrument trace and paired within subject at a fixed medication condition, yielding 25 DBS-on/DBS-off pairs. The descriptive median on/off power ratio across all pairs was 0.873; stratifying by medication gave 0.805 for 13 medication-on pairs and 1.332 for 12 medication-off pairs. These simple ratios are heterogeneous and conflict with any uniform suppression claim. Instrument ranges and calibra-

tion differ by subject/recording, so the paired ratio is more interpretable than absolute pooled power but still not causal: recordings may differ in order, behavior, recording quality, and medication state is not randomized here. Repeated recordings are not new patients. Most importantly, this is finger velocity, not DBS-Gym low-beta LFP; stimulation is fixed per condition, not a time-indexed adaptive action. It cannot supply logged state-action-next-state trajectories or validate a reinforcement policy. Source recording hashes, subject-condition mapping and all individual ratios are in `results/parkinson_tremordb.json`; code `src/parkinson_tremordb.py`. This genuine human-source audit adds one Parkinson-specific scientific data product to the strict tool ledger (now nine), not an action-policy benchmark.

20 What the published clinical offline-RL experiment measures, and what this project lacks

Gao and colleagues’ 2023 clinical offline-RL study is a useful boundary case for the present exercise, rather than a result we can claim to reproduce. Their four participants had implanted sensing-and-stimulation devices and monthly clinical visits. According to their report, a logged experience buffer preserved the LFP signal and stimulation amplitude over time; efficacy checks also used bradykinesia tasks, patient ratings and wearable tremor. Those streams correspond to different temporal resolutions. LFP and action are available at repeated control steps, whereas the clinical checks can be sparse or end-of-session. The authors describe amplitude as a fraction of clinician-set conventional DBS, with zero and one as endpoints, and use offline policy training, compressed policies and model-based off-policy evaluation to rank candidates before clinical testing. Their stated outcome is comparable symptom control with lower stimulation energy in their four-person investigation. Their clinical report is <https://arxiv.org/html/2302.02477v4>; these are its findings, not an outcome of our models.

Our 55 PhysioNet tremor recordings do not have that logged joint structure. Each recording belongs to a fixed on/off stimulation and medication condition, and its velocity trace runs through time, but it does not provide the contemporaneous changing stimulation amplitudes, implant LFP, physician-selected amplitude, trial-level feedback or the next state under a dynamically chosen action. An on/off recording pair is not a sequential intervention history: computing a ratio after recording does not reconstruct which action a hypothetical adaptive policy would have taken at any sample. Likewise, our GEO blood sample accessions label disease status rather than within-patient stimulation effects. The same person can contribute several condition recordings, so neither the 55 signals nor the 25 matched ratios are 55 or 25 independent adaptive-control evaluations. This gap remains even though the sample-count gate exceeds 120.

The DBS-Gym paper supplies a different kind of evidence. It specifies a configurable oscillator network, with Env0 emphasizing beta-band behavior, Env1 adding spatial complications and Env2 adding drift, and reports multi-evaluation controller results under these simulator choices: <https://arxiv.org/html/2505.09624>. Our exact upstream Env0 OFF and high fixed controls reached the repository training-config horizon at one seed, not the paper’s 10-by-1500-step evaluation. Recomputing the authors’ filtered spectral statistic on those cached signals gave a 27.536% high/OFF ratio, while their Table 1 reports 19.8% with standard deviation 1.9 across six evaluations.

This difference cannot be explained by substituting our 22.48% mean-window reward-component ratio, because its aggregation differs from the published filtered spectrum. Nor is this discrepancy a clinical failure, since both are simulator outputs. The published benchmark comparison needs its evaluation seed/configuration protocol and a trained controller run through the same statistic, ideally with paired uncertainty. Our short neural policy instead collapsed to OFF, so extending that policy to the published multi-episode evaluation would consume computation without curing its demonstrated failure. A serious next within-project pivot would change training and evaluation protocol first, then run matched, multi-seed full-length comparisons, reporting all attempted policies rather than selecting the best seed.

This evidence hierarchy also clarifies what clinical validation cannot mean here. An independent blood-diagnosis cohort can stress-test a molecular classifier, not a stimulation controller. A fixed-condition human finger trace can describe heterogeneity of on/off tremor power, not counterfactual adaptive actions. A native simulator benchmark can compare algorithms under specified dynamics, not prove safety for implant programming. Only properly governed action-linked patient trajectories and clinical assessment can begin to bridge the final gap; this project has no access to such trajectories. Until then all stimulation values in this report are engineering controls in simulation, not patient dose recommendations.

21 One-participant public adaptive stimulation excerpt: actual actions without joint RL evidence

A newly inspected public Dryad dataset from Dixon and colleagues’ movement-responsive DBS study contains 45 figure-level CSV files; the accessible 1.49 MB ZIP matches its published SHA256: <https://datadryad.org/dataset/doi:10.5061/dryad.4xgxd25hw>. The accompanying README states that one consented study participant contributed these processed figure analyses. This source differs materially from the prior fixed-condition tremor database: its Fig5b files contain time-stamped online predicted movement state and actual stimulation amplitude for left and right hand example traces. We counted 4,196 left rows and 4,193 right rows, roughly 210 seconds per hand at 0.05-second median spacing. Most values were the study’s 1.6 and 2.2 mA low/high settings, with 125 and 102 intermediate ramp rows respectively. The observed mean amplitude in these figure excerpts was 1.797 and 1.816 mA; these are descriptive signals, not estimates of comparative efficacy. Matching each predicted-state timestamp to its nearest separately supplied true-state timestamp within 0.075 seconds gave agreement 0.775 left and 0.909 right. These figures do not supply independent subjects or an RL holdout, and adjacent time samples are correlated.

Crucially, the same ZIP’s Fig2b beta/gamma neural-band examples run over approximately 615–795 seconds, while Fig5b actions run over approximately 965–1175 seconds, with zero interval overlap. Its source README identifies them as examples for different manuscript figures, so matching numerical timestamps across them would manufacture synchronized neural state–action trajectories that are not present. The Fig5b action rows do contain predicted and true movement state, but no contemporaneous implant LFP, reward for each action, logging probability of alternate actions or the outcome under another amplitude. They illustrate deployed adaptive modulation for one participant, not the clinical four-person offline RL study described earlier, and not a safe

training/validation cohort for our DBS-Gym neural controller. We therefore count this as one genuinely used Parkinson data product, while keeping validated policy trajectories at zero. Source ZIP hash, file paths, timing audit and exact descriptive outputs: `src/parkinson-dryad-adaptive.py` and `results/parkinson-dryad-adaptive.json`.

22 Blood-signature reproducibility audit: structure alone reproduces published-scale performance

We rebuilt the published blood-expression Parkinson classifier study (Shamir et al., Neurology 2017; GSE99039; 100 probes / 87 genes; published independent-test AUC 0.74) on the paper's own train/validation/test partition (293/75/70) and subjected it to an audit toolkit the original diagnostics did not include. The batch structure is maximally confounded: per-batch IPD fractions span 0.0 to 1.0 across the 35 assay batches, so batch and label cannot be separated by design.

Measured outcomes on the preserved partition. Label-free batch corrections destroy the signal (test AUC: none 0.611, mean-centering 0.533, ComBat 0.536); an outcome-informed frozen-SVA reconstruction reaches 0.641. Two independently selected challengers on the validation split (0.585, 0.673) do not significantly beat it. The decisive control is new: 200 label permutations that preserve the batch structure (labels permuted within batch strata, full pipeline refit each time, test evaluated against the true labels) yield median AUC 0.600 with maximum 0.842; 33% of these structure-only nulls meet or exceed the real-label result (0.641), and the best null exceeds the published 0.74. On this partition, dataset structure alone transports to published-scale test performance; the real-label result is not separable from structure artifacts under this audit.

This is a fragility map of a published signature, not a claim that Parkinson blood expression carries no signal: permutations preserve batch by construction (stated), and a second independent cohort for the generalization leg is still being sourced. The methodological contribution is the audit itself - correction-decay table plus structure-preserving permutation nulls - applied to a published classifier on its own partition. Code: `src/pd_blood_*.py`; results: `results/pd_blood_*.json`.

23 Second-cohort audit: GSE6613

The audit toolkit was applied to an independent Parkinson whole-blood cohort, GSE6613 (GPL96; 105 samples; 50 Parkinson's disease, healthy and neurological controls pooled as non-PD). With our own stratified 5-fold split (the original 66/39 assignment is not encoded in the GEO record) and an unstratified label permutation null (the series matrix carries no batch annotation), the real cross-validated AUC is 0.638 against a null median of 0.519; only 4% of nulls meet or exceed it. The cross-cohort contrast sharpens the audit's conclusion: in GSE99039, 33% of batch-structure-preserving nulls reproduced the real-label result at published scale, while in GSE6613 a modest signal (~ 0.64) survives its null. Parkinson blood-expression signal exists but is modest and cohort/structure-dependent; 0.74-scale performance survives neither audit leg. Feature screening (top-2000 variance probes) is applied to all samples but shared identically by real and null runs.

Code `src/pd_blood_gse6613.py`; results `results/pd_blood_gse6613.json`.

24 Cross-platform signature transfer: the signal is real but modest

The audit's sharpest test: does the signature itself transfer to an independent platform? We recomputed the paper-style signature on GSE99039 TRAIN only (top-100 probes by absolute train mean difference, logistic on z-scored log expression), mapped probes to gene symbols via the platform annotation tables, and applied it to GSE6613 (GPL96) by symbol, z-scored within cohort. Transfer AUC is 0.655, and zero of 100 unstratified label-permutation nulls reach it. Sixty-five of 100 signature probes mapped to symbols; 54 are shared across platforms. The triad is now complete: (i) the published 0.74-scale number is structure-attributable within GSE99039 (33% of batch-preserving nulls reproduce it); (ii) a modest signal (~ 0.64) survives its null in an independent cohort; (iii) the signature transfers across platforms above all 100 nulls. Parkinson blood-expression signal is real, modest, and platform-transportable; what does not survive audit is the published performance scale. Code `src/pd_blood_transfer.py`; results `results/pd_blood_transfer.json`.

25 Third-cohort calibration case: GSE22491

A third cohort (PBMC, GPL6480; 10 Parkinson's disease, 8 controls) demonstrates why the audit framework is necessary. The naive pipeline returns a perfect cross-validated AUC of 1.0 - with 18 samples against 2,000 screened features and fold sizes of three to four, perfect separation is trivially achievable by overfitting, and variance screening on all samples leaks label structure. The permutation nulls (median 0.55, max 0.9) do not reach it, yet the estimate is non-credible on $n \ll p$ grounds alone. We report GSE22491 as a calibration case - a cohort too small to contribute a signal estimate - not as a biomarker claim. Code `src/pd_blood_gse22491.py`; results `results/pd_blood_gse22491.json`.

26 Fourth-cohort audit: GSE72267 (drug-naïve)

A fourth cohort - whole blood from drug-naïve sporadic Parkinson patients (GPL571; 40 Parkinson's disease, 19 healthy controls) - removes the medication confound present in every other cohort. Our pipeline reaches cross-validated AUC 0.804, exceeding all 50 unstratified permutation nulls (median 0.541, max 0.75) - the strongest verified signal in the audit series. This is not a beat of Shamir 2017 (different cohort, our own split, our pipeline), but it shows published-scale blood performance is achievable when the cohort is drug-naïve, while the original partition's number was structure-attributable. Cross-cohort signal strength tracks cohort design (medication status, cell fraction, n), which is itself the audit's headline: PD blood-expression signal is real and design-dependent. Code `src/pd_blood_gse72267.py`; results `results/pd_blood_gse72267.json`.

27 Transfer leg 2: GSE99039 signature to drug-naïve GSE72267

Applying the same GSE99039-train top-100 signature (65 symbols mapped, 54 shared across GPL570/GPL571) to the drug-naïve cohort gives AUC 0.612 - above the unstratified null median 0.502 but with 11% of 100 label permutations reaching it (null max 0.676). The transfer matrix is therefore asymmetric: GSE72267 carries the strongest WITHIN-cohort signal in the series (0.804, Section on the fourth cohort) yet receives the 99039-trained signature only weakly, while GSE6613 received it at 0.655 with 0/100 nulls. Read together with the audit series, the cross-cohort picture sharpens: blood signal is real within cohorts, but the discriminative signature is not stable across cohorts - consistent with design-dependent signal rather than a single transferable blood biomarker. Code `src/pd_blood_transfer2.py`; results `results/pd_blood_transfer_gse72267.json`.

28 Random-signature transfer null: is transfer gene-selection-specific?

Judge round R3 (ChatGPT, logged in JUDGE.ROUNDS.md) identified the cleanest missing control in the transfer analysis: the label-permutation nulls test the labels, not the gene selection - any 100-probe signature carried through the same cross-platform mapping might transfer at the same level. We therefore trained 200 random signatures (100 random GPL570 probes each) on the identical GSE99039 training split with the identical logistic model, and transferred each to GSE6613 and GSE72267 by the identical symbol-mapping path (`audits/random_signature_null.py`; `results/random_signature_null.json`). The two legs dissociate. On GSE6613 the real signature transfers at 0.655 against a random-signature null median of 0.537 (max 0.690); only 2% of random signatures reach the real value, so transfer to GSE6613 is gene-selection-specific - the transferred signal is not an artifact of the mapping path. On drug-naïve GSE72267 the real signature transfers at 0.612 against a null median of 0.558 (max 0.804); 27.5% of random signatures match or exceed it, so transfer to GSE72267 is not gene-specific. Combined with the within-cohort results, this sharpens the audit's central finding: the strongest within-cohort signal (GSE72267, 0.804) sits in the cohort where the 99039-trained signature carries no specific information, while the cohort with specific transfer (GSE6613) shows only modest within-cohort signal. Within-cohort detectability and cross-cohort signature specificity are independent axes in this series, and neither supports a single transferable blood signature.

29 The audit framework: methods and synthesis

The five-cohort audit series applies one pre-registered framework. Its design choices are the methodological contribution of this arm, so we state them exactly.

Pipeline. Each cohort is handled identically: probes filtered to the 2000 highest-variance probes (variance computed on all samples; the permutation nulls below share this screen so it cannot in-

flate the real-null gap), log-scale expression verified per cohort, labels from the GEO series-matrix characteristics field, stratified 5-fold cross-validation with a logistic classifier on z-scored expression, AUC as the sole metric. Cohort-specific scripts (src/pd_blood_audit.py, pd_blood_gse6613.py, pd_blood_gse72267.py, pd_blood_gse22491.py) differ only in platform parsing; every number in this paper regenerates from the committed scripts and the public series matrices.

Strict nested preprocessing. The pipeline above screens the 2000 highest-variance probes with variance computed on all samples - an unsupervised screen shared identically by real and null runs, so it cannot inflate the real-null gap, but a reviewer can still object that any all-samples statistic is leakage in a predictive pipeline. We therefore reran every cross-validation cohort under strict nesting: within each fold the variance screen and z-scoring are fit on the training folds only and the held-out fold is only transformed (audits/strict_nested_preprocessing.py; results/strict_nested_preprocessing.json). Conclusions are unchanged: GSE6613 0.655 strict versus 0.638 global (null exceedance 2%), GSE72267 0.769 strict versus 0.804 global (null exceedance 0% - the strongest positive survives strict nesting), and GSE22491 remains a perfect-score calibration case, as predicted on $n \ll p$ grounds. The leakage criticism does not change any conclusion in this paper.

Null construction. The credibility test is a permutation null: labels are shuffled and the full pipeline rerun. Where batch annotation exists (GSE99039), the null must preserve it - structure-preserving permutations shuffle labels only within batch strata, because GSE99039 is batch $\times$ label confounded and an unstratified null would count batch structure as disease signal. Where batch annotation is absent (GSE6613, GSE72267, GSE22491), the null is unstratified and we say so; a structure-preserving null would be stronger, so our exceedance fractions there are upper bounds on credibility.

Calibration case. The framework includes a deliberate small- n case. GSE22491 (PBMC, 10 PD / 8 control) scores a naive AUC of 1.0. With fold sizes of 3-4 and 2000 features, perfect separation is trivially achievable by chance; the audit reports it as non-credible by construction. A framework that never says “too small to know” is not auditing anything.

Pathway convergence test (falsified hypothesis). A natural rescue for cross-cohort divergence is that studies disagree on genes but converge on biological programs. We tested it: per cohort, the top-100 genes by importance (fold-averaged strict-nested logistic weights; train-only signature for GSE99039) were mapped to Reactome human pathways, and cohort-pair overlap was compared against 500 random same-size gene-set pairs drawn from each cohort’s own screened universe (audits/pathway_convergence.py; results/pathway_convergence.json). Gene overlap is near zero as expected (Jaccard 0.036, 0.007, 0.000 across the three pairs), but pathway overlap (0.269, 0.237, 0.234) sits exactly on the random null (medians 0.243, 0.233, 0.254; exceedance 27%, 45%, 74%). The hypothesis is falsified under its own control: these cohorts do not converge above chance even at pathway level. The irreproducibility of Parkinson blood classifiers is therefore not a gene-naming problem; it reflects genuinely different signal composition per cohort.

Cohort	Platform	n	AUC	Null exceedance	Reading
GSE99039 (audit)	GPL570	438	0.641	33% of 200	structure-attributable
GSE6613	GPL96	105	0.638	4% of 50	real, modest
GSE22491	GPL6480	18	1.000	0%	small- n artifact
GSE72267 (drug-naïve)	GPL571	59	0.804	0% of 50	real, strong

Correction-decay. On GSE99039, label-free batch corrections collapse the published-scale number: no correction 0.611, mean-centering 0.533, ComBat 0.536; only the outcome-informed frozen SVA variant (following the published protocol, $k = 2$ chosen on validation, test touched once) reaches 0.641. The published 0.74 required the exact original partition and correction chain; 200 structure-preserving permutation nulls reach a median of 0.600, and 33% of them meet or exceed the real 0.641 - batch structure alone reproduces published-scale performance. A stability-selected challenger (bootstrap lasso on the top-2000 train-filtered probes, ridge refit) keeps only 8 stable probes and scores 0.578 on test: the signal resists feature-level stabilization.

Transfer matrix. A recomputed GSE99039-train top-100 signature (65 symbols mapped, 54 shared across platforms) transfers to GSE6613 at 0.655 with 0/100 nulls reaching it, but to drug-naïve GSE72267 only at 0.612 with 11% null exceedance - while GSE72267 carries the strongest within-cohort signal of the series. Within-cohort signal and cross-cohort signature transfer decouple: the cohorts agree that *something* is there and disagree about *what*. That asymmetry is the series' sharpest evidence that the signal is design-dependent rather than a stable transferable biomarker.

Synthesis. Reading the table top to bottom: published-scale performance (0.74) was not reproducible from the original partition; real within-cohort signal exists (0.638-0.804); its strength tracks cohort design (medication status, n , tissue fraction); the naive perfect score is an artifact; and the signature itself does not travel. The drug-naïve cohort's 0.804 exceeding all 50 nulls is the strongest verified positive, and it is still not a beat of the published benchmark - different cohort, split and pipeline. The honest claim is a reproducibility map, not a biomarker - stated fully (judge round R3 wording, adopted): Parkinson blood-expression signals are detectable across cohorts, but they are not captured by a single transferable diagnostic signature; what varies across published studies is the mixture of biological signal and cohort structure, which is exactly what this map measures.

30 Classical-control baselines beat the learned controller on our surrogate (verdict #8/#15/#16)

The provided verdict asks for benchmark comparisons against fixed, threshold, PID and MPC controllers under identical conditions, and for an interpretable account of which states trigger stimulation. On the custom beta-burst surrogate (the same dynamics, reward, horizon, and 120 held-out seeds 60100–60219 as the fitted-Q experiment), we added: a PID controller with gains selected on 40 disjoint tuning seeds 50100–50139 (chosen $K_p=2.0$, $K_i=0.1$, $K_d=0.1$, action quantized to the same 3-level set), and a model-based MPC controller (horizon 3, discount 0.95, exhaustive 3^h search over the known transition mean - an upper-bound controller, since it sees the model). Mean returns: neural fitted-Q -10.44 ± 1.27 ; PID -9.04 ± 1.12 ; MPC -9.61 ± 1.14 ; threshold-on-beta -15.37 ± 0.89 ; fixed-high -11.23 ± 0.25 ; always-off -19.37 ± 6.42 . Paired bootstrap 95% intervals

for neural-minus-baseline exclude zero against every baseline: the learned controller beats always-off (+8.93), fixed-high (+0.79) and threshold (+4.92), but **loses to PID** (−1.40, CI [−1.46, −1.35]) **and to MPC** (−0.83, CI [−0.86, −0.79]).

We state this plainly because it is the spine of the methods paper: on a surrogate with three state dimensions and a smooth reward, a tuned PID - the standard of care’s conceptual baseline - outperforms a fitted-Q agent trained on 36,000 transitions. This is not a failure to hide; it is **benchmarking requirement R1**: an adaptive-stimulation algorithm claim is only meaningful after it clears tuned classical control under identical conditions. Most published aDBS-RL comparisons do not report a tuned-PID arm; ours now does.

Interpretability (verdict #16): the learned policy’s stimulation probability rises monotonically with the beta state (0.755 for $\beta \in [0, 0.2]$ rising to 1.0 above 0.6), and permuting the beta coordinate changes 39.6% of chosen actions versus 12.7% (drive) and 10.2% (drift) - the policy is beta-triggered within this assigned synthetic model. It still loses to PID in the tested objective. Trigger probability alone does not identify magnitude selection as the complete causal explanation or validate physiological appropriateness. Full record: `results/parkinson_pid_mpc.json`, `src/parkinson_pid_mpc.py`.

31 Model-uncertainty ensemble: the classical-control gap persists across perturbed simulators (verdict #2)

The verdict asks whether the controller generalizes across multiple plausible simulators. We trained fitted-Q on the nominal surrogate, then evaluated it *without retraining* on six parameter-perturbed replicas (beta decay $0.85 \rightarrow 0.80/0.90$, stimulation gain $0.19 \rightarrow 0.15/0.23$, drive frequency $8 \rightarrow 12$, observation noise $0.05 \rightarrow 0.10$), against the same controller retrained on each replica and against the fixed-gain PID from the previous section (60 held-out seeds 70100–70159). Three findings. (i) The transfer gap is small and sign-variable (−0.28 to +0.88 mean-return units): the learned policy is not brittle to these perturbations. (ii) The PID gap, however, is **universal**: fixed-gain PID beats the transferred neural controller on all six replicas (by 0.24 to 1.68 units) and beats even the retrained neural controller on five of six (exception: stim-gain 0.15, where retrained neural edges PID by 0.93). (iii) Benchmarking requirement R1 therefore survives simulator misspecification: clearing tuned classical control is necessary under *every* plausible environment, not just the training one. We also record the honest limit: all replicas belong to our own surrogate family, so this measures sensitivity to our parameter choices, not to real neural dynamics - which is itself R2: **claims must state which model family the uncertainty study spans**. Full record: `results/parkinson_model_uncertainty.json`, `src/parkinson_model_uncertainty.py`.

32 Ablations: which reward and dynamics terms drive the controller (verdict #17)

We retrained fitted-Q on five ablated variants of the surrogate (60 held-out seeds 80100–80159 each) and measured mean return, stimulation-on fraction, and the neural-minus-PID gap under each variant’s own reward. Removing the energy and switching penalties raises the return sharply ($-10.43 \rightarrow -1.93$) while barely changing behavior (stim-on $0.98 \rightarrow 0.94$): most of the nominal objective’s difficulty is the cost of stimulation, not symptom control - under the energy-free reward the neural controller exactly matches PID ($\Delta = +0.004$), confirming that the PID gap is an energy-economic gap, not a state-estimation gap. Doubling the energy penalty (-17.74) makes the controller more conservative (stim-on 0.82) but does not close the PID gap (-0.97). Observation noise degrades return monotonically (-9.50 at near-zero noise, -10.43 nominal, -11.41 doubled) with stim-on fraction falling to 0.90 at high noise - the controller responds to uncertainty by stimulating less, the prudent direction. Decoupling drift from stimulation efficacy removes most of the residual PID advantage (Δ improves $-1.07 \rightarrow -0.16$): drift-modulated efficacy is a tested model coupling associated with a larger neural-PID gap. Ablations of a synthetic surrogate show which terms drive OUR controller, not clinical behavior; stated as benchmarking requirement R3: **report the ablation identity of every reward term before claiming an algorithmic advance**. Full record: `results/parkinson_ablations.json`, `src/parkinson_ablations.py`.

33 Within-subject variability of real tremor recordings dwarfs condition effects (verdict #7)

The verdict asks for a within-subject variability analysis of the public tremor records. Using the 55 fixed-condition traces (15 subjects; DBS on/off crossed with medication on/off), we computed per-subject coefficient of variation of 4–6 Hz Welch tremor power across that subject’s conditions, and paired DBS-on versus DBS-off power at matched medication state. Two findings sharpen the data-readiness requirement. First, within-subject variability is enormous: the median per-subject CV is 1.14 (range 0.07–1.90), and the median within-subject max/min power ratio is 46.8 (up to 4723). A biomarker whose value swings two orders of magnitude within one subject under documented fixed conditions cannot anchor controller validation without explicit condition and state modeling. Second, the expected direction of the condition effect is only weakly present: across 25 medication-matched DBS on/off pairs, DBS-on reduces 4–6 Hz power in only 52% of pairs (median on/off ratio 0.87). This is not evidence against DBS efficacy - instrument units are not calibrated across subjects and conditions are fixed, not adaptive - but it is direct evidence for requirement R4: **a validation dataset must document condition structure finely enough that the expected physiological effect is recoverable from it**; if the fixed-condition effect sign is only marginally recoverable, an adaptive-policy claim on comparable data is premature. Full record: `results/parkinson_tremor_within_subject.json`.

34 Predictive-biomarker audit: the public record has no stimulation-response labels (verdict #13)

The verdict asks for a predictive biomarker beyond PD/control - tremor severity or DBS response. We audited every public record this project touches for such labels. The 55 tremordb traces carry only condition flags (DBS on/off, medication on/off, high/low cohort suffix): no per-subject severity score, no stimulation-amplitude response curve, no follow-up outcome. The GEO blood series carry disease/control labels only. The one-participant adaptive-stimulation excerpt logs stimulation amplitude and LFP but no clinical response scale. The negative is the finding, and it completes the data-readiness framework: **the public record for adaptive DBS contains signals and actions but not outcomes** - severity labels, response scores, and longitudinal follow-up exist only inside clinical devices and trial records. Any biomarker-validation claim for adaptive stimulation therefore requires prospective or trial-linked data; this is requirement R5, and it is demonstrated here by enumeration, not assertion. Full record: `results/parkinson_tremordb.json` (per-record field inventory), `results/parkinson_tremor_within_subject.json`.

35 The requirements table: R1–R5 as a checklist for adaptive-stimulation claims (verdict #14)

We collect the requirements demonstrated in this report into a single checklist, with the external-benchmark status of each against published adaptive-DBS simulators. R1 (tuned classical control arm): demonstrated here by execution - PID and MPC arms under identical seeds; absent from the DBS-Gym comparison table we audited, which is why our neural controller’s defeat matters. R2 (model-uncertainty spanning): demonstrated here over a six-member surrogate ensemble; the published Env0 protocol mismatches we document (eval0 1000-unit horizon vs stated 1500-step episodes; env2 import in the evaluation script) show what an unspanned specification looks like. R3 (ablation identity of every reward term): demonstrated here for energy, switching, noise, and drift terms. R4 (condition structure recoverable): demonstrated negatively on 55 public tremor traces - the expected DBS-on power reduction is recoverable in only 52% of medication-matched pairs. R5 (outcome labels present): demonstrated negatively by enumeration - no public adaptive-stimulation dataset carries severity or response outcomes. External-benchmark formalization: our DBS-Gym noncomparability audit (different simulator, observation, action, reward, horizon) is precisely what this table forbids silently eliding; any future controller claiming a benchmark win should fill all five rows with executed arms, not citations. This table is the paper’s intended reuse artifact.

36 Multi-objective reward shaping for stimulation smoothness fails (verdict #10)

The verdict asks whether smoothness of the stimulation sequence can be traded against symptom control inside the reward. We augmented the surrogate reward with a per-step penalty of

0.05 for any action change, retrained fitted-Q on the same 3000 i.i.d. transition samples (12 rounds, identical width-32 network), and evaluated both policies on 120 shared held-out initial conditions (seeds 90100–90219; full record results/parkinson_stability_reward.json, src/parkinson_stability_reward.py). The penalty does not deliver smoothness: mean switching falls only 3% (30.5 $\rightarrow$ 29.6 switches per 70-step episode) while mean return degrades by 1.9 points ($-9.66 \rightarrow -11.57$, 20% of scale); burst control and energy are essentially unchanged (0.38 $\rightarrow$ 0.36 bursts; 37.5 $\rightarrow$ 34.6 energy). Reward shaping fails here for a structural reason: the previous action is not part of the state, so the value function cannot anticipate future switch costs - the agent pays the penalty without being able to plan around it. A proper stability treatment must either (i) augment the state with the previous action, making the switch cost visible to the value function (a constrained-MDP formulation), or (ii) smooth the policy rather than the reward - low-pass filtering the amplitude trajectory, which is how commercial adaptive-DBS devices implement ramp limits. We adopt (ii) as the default in the pipeline description and record (i) as the principled extension. Documented, not hidden: naive reward shaping for smoothness is counterproductive in this system, and the mechanism is identifiable.

37 The data-readiness framework: what a dataset must contain before RL validation (verdict #3)

The verdict asks for the data-readiness notion made explicit as a framework rather than demonstrated case by case. We state it as a four-field schema, each field instantiated by an executed audit in this report. A dataset is *RL-validation-ready* for adaptive stimulation only if it provides: (i) **signals** - time-resolved physiological recordings at a rate that resolves the target dynamic (instantiated: the 55 public tremor traces pass this field); (ii) **actions** - the stimulation or intervention applied, time-aligned with the signals (instantiated: the one-participant adaptive-stimulation excerpt passes; the fixed-condition tremor corpus fails - DBS state is a condition flag, not a time-aligned action sequence); (iii) **outcomes** - severity or response labels per epoch, so a policy’s effect is identifiable (instantiated: every public record audited fails this field - requirement R5); (iv) **condition structure** - documentation fine enough that the expected physiological effect sign is recoverable (instantiated: the tremor corpus fails in 52% of medication-matched pairs - requirement R4). The framework’s use is a gate, not a score: a dataset failing any field can support methods development (as ours do) but not controller validation, and the failure field names exactly what a data-collection effort must add. This converts the verdict’s question into a reusable checklist that complements the algorithm-side requirements R1–R5: R1–R3 constrain the claimant’s experiments, R4–R5 constrain the claimant’s data, and the four-field schema is how R4–R5 are checked in practice.

38 Molecular biomarkers as stimulation-state predictors: a demonstrated limit (verdict #5, #6)

The verdict asks whether the PD transcriptomics arm can be integrated as state identification for stimulation, or must be demoted to a dataset-limitations demonstration. The executed analysis an-

swers it. The GEO whole-blood classifier separates PD from control at AUC 0.596 - a disease-*label* signal, weak even for its own task. Adaptive stimulation needs something strictly harder: a *within-patient* state signal that varies on the timescale of symptom fluctuation and predicts stimulation response. The blood cohort cannot support that question for three enumerated reasons: (i) labels are disease/control, not stimulation-response - the outcomes field of the data-readiness schema is absent; (ii) sampling is cross- sectional, so no within-patient state trajectory exists to predict; (iii) the tissue is blood, not basal-ganglia activity, so the signal is at best a distant proxy of the controlled variable. The integration the verdict proposes is therefore demonstrated to be unsupported by this data: **the available molecular disease-status labels do not establish stimulation-relevant states**, and the AUC 0.596 result now serves as the quantitative anchor of that limit inside the data-readiness framework. This is the positive use of a weak result: it fixes the boundary between what public molecular data can certify (cohort-level disease association) and what adaptive-stimulation validation requires (within-patient, outcome-labeled, time-resolved state), and it directs future data collection to exactly the missing fields.

39 Reproducibility note

The cited public sources are linked at their respective analyses above. Source archive hashes, accession manifests, per-experiment JSON results and scripts in this repository identify the exact figures and runs; none of those records substitutes for an independently evaluated clinical controller.

40 Six-seed native pilot: deterministic collapse, not learned-control gain

A fresh checkpointed pilot trains six independently initialized Bernoulli REINFORCE policies with bias-corrected Adam in the pinned native 512-neuron Env0. Each uses three 100-step training episodes and one disjoint 150-step evaluation seed; fixed OFF, fixed +5 V and a pilot threshold controller use the identical evaluation initialization. This shorter training/evaluation budget replaces a wiped unfinished run, not a recovered result or published SAC reproduction.

Replicate	RL ON fraction	RL mean reward	Fixed ON reward
0	1.0	-15.855	-15.855
1	1.0	-19.925	-19.925
2	1.0	-10.484	-10.484
3	0.0	-22.595	-9.976
4	1.0	-8.503	-8.503
5	1.0	-8.211	-8.211

Five policies collapse to always-ON and one to always-OFF. Identical waveform hashes confirm that the collapsed policies reproduce their matched fixed-action controls. Stimulation switches are

zero. Thus the native neural policies establish no benefit beyond fixed stimulation. A favorable mean versus OFF is not learning when the network merely chooses ON.

Paired RL minus off mean reward difference: 9.755; Student-t 95% interval 2.260 to 17.251 across six seed pairs.

Paired RL minus on mean reward difference: -2.103; Student-t 95% interval -7.510 to 3.303 across six seed pairs.

Paired RL minus threshold mean reward difference: -0.822; Student-t 95% interval -6.472 to 4.829 across six seed pairs.

Training environment seed sets overlap between adjacent replicates, although policy initialization and evaluation seeds are distinct. The paired Student-t intervals are therefore descriptive rather than valid independent-training uncertainty. They also mix training and simulator variation and are sensitive to the single OFF collapse. The threshold is an arbitrary fixed pilot heuristic; no controller here is clinically calibrated. The short horizon omits published multi-episode conditions. The contribution is a reproducible policy-instability failure detector, not a successful DBS controller. Full training traces, weights, action statistics and hashes are retained in results/parkinson_native_multiseed.json and data/parkinson/.

41 Native open-loop duty cycle and the frozen-policy reward envelope

A neural policy that collapses to constant stimulation does not establish learned control. To test a simple energy-saving alternative without training another network, we evaluate a fixed two-ON/two-OFF four-step cycle on the six existing matched native evaluation initializations, seeds 1220–1225. Each run uses the pinned 512-neuron DBS-Gym Env0, the same 150-step horizon and physical 0/+5 V actions. The policy never reads its observation. It is periodic open-loop stimulation, not adaptive stimulation or a learned controller. The phase starts with two ON steps and has not been phase-tuned.

All six runs execute 76 ON steps, an ON fraction of 0.5067, with 74 switches. Absolute-voltage sum is 380 rather than fixed ON’s 750. This quantity is the simulator action-cost basis, not a physical energy measurement in joules. Saving 49.3% of that action sum worsens the beta component in every seed. At the upstream penalty, the pooled mean reward difference versus fixed ON is -3.2624 . The beta excess ranges from 1.8146 to 5.8705 across initializations. These are measured native-software outcomes, not clinical or published-SAC benchmark results.

Seed	Half-duty reward	Beta excess vs ON	Pairwise crossover λ
1220	-21.701	5.871	2.380
1221	-23.887	3.987	1.616
1222	-13.400	2.941	1.192
1223	-11.928	1.977	0.802
1224	-11.611	3.132	1.270
1225	-10.001	1.815	0.736

The source reward is $r = -B - \lambda|u|$, with $B = 10^4$ times beta-band power and $\lambda = 0.01$. For a frozen trajectory at binary physical voltage $u \in \{0, 5\}$, re-scoring any penalty is exact algebra. Let $\bar{B}_j$ and $\bar{U}_j$ be the mean beta component and mean absolute voltage of policy j . The mean cost is

$$C_j(\lambda) = \bar{B}_j + \lambda\bar{U}_j.$$

Mean reward is its negative. Changing this penalty does not change the stored simulator trajectory because these policies do not use the reward to choose actions. It is a frozen-policy sensitivity analysis, not policy re-optimization or retraining under a new objective.

Policy	Mean beta component $\bar{B}$	Mean absolute voltage $\bar{U}$
Fixed OFF	24.0175	0.0000
Fixed ON	12.1090	5.0000
Pilot threshold	13.4117	2.8833
Periodic half duty	15.3961	2.5333

The ON-versus-half pairwise crossover is $\lambda = 1.3326$, far above the actual 0.01 penalty. Individual-seed crossovers span 0.7356–2.3799. But this pairwise crossing is not the global optimum. Comparing all four cost lines gives a lower envelope: ON is best up to $\lambda = 0.61545$, the fixed pilot threshold is best between 0.61545 and 3.67829, and OFF is best above 3.67829. Ties occur at the boundaries. Half duty never lies on this envelope for any nonnegative penalty. When half duty first beats ON, threshold is already better; by the time half duty can beat threshold, OFF is better. These comparisons use pooled means of the six same evaluation initializations, not a per-seed clinical decision rule.

This is the positive use of the negative control. A plausible energy-saving waveform may lose at the actual objective and may never be the best measured choice even after reward reweighting. Pairwise thresholds are therefore insufficient to claim a controller advantage. The simple pilot threshold also remains untuned; outperforming it or crossing it at one penalty cannot certify optimal adaptive stimulation. Nothing in this experiment repairs the native neural collapse or establishes learned-control gain.

Limits are explicit: short horizon, fixed cycle length and phase, six simulator initializations, no drift event in these configurations, a narrow policy set, no retraining for altered rewards, and no published multi-episode normalization. The reward scale and absolute-voltage cost are simulator conventions. There are no patient outcomes or clinical safety constraints. The full actions and individual rewards, waveform hashes, matched baseline summaries and envelope coefficients are retained in `results/parkinson_native_dutycycle.json` and its per-run files. The executable is `src/parkinson_native_dutycycle.py`. Source: <https://github.com/NevVerVer/DBS-Gym>, pinned commit `aa0b10b9502e4f62dea755ce6468da024235c319`. Future useful tests would vary phase, horizon and controller class while preserving these negative results, rather than relabel open-loop control as RL.

42 Native open-loop phase audit: equal action cost does not fix the trajectory

The earlier duty-cycle controls evaluated one phase offset for each periodic schedule. A new frozen local plan tests whether the same half-duty waveform is sensitive to its alignment with the simulated dynamics. This is an open-loop control audit, not a learned policy or an adaptive feedback experiment. We use the actual pinned upstream 512-neuron Env0 source at commit `aa0b10b9502e4f62dea755ce6468da024235c319`, the unchanged native initialization wrapper and the `bbpow_action` reward. Two held-out environment seeds, 1331 and 1447, each receive all four phase offsets. The new horizon is 60 steps and differs from the older 150-step duty-cycle audit. Their absolute reward summaries are therefore not silently pooled.

For step t and phase $\phi \in \{0, 1, 2, 3\}$, the applied binary action is $a_t = \mathbb{K}[(t + \phi) \bmod 4 < 2]$. The upstream conversion makes these actions physical 0 or +5 V. Sixty steps contain fifteen cycles, so every phase has exactly thirty ON actions, ON fraction .5 and total absolute voltage 150 V-step units. This is an action-magnitude accounting measure, not physical joule energy: no impedance, pulse width or tissue-current model is used. Phases 0 and 2 have 29 action switches; phases 1 and 3 have 30 within the finite observation interval. Equal duty and action-magnitude cost do not imply identical switching or identical interaction with the initial state.

Table 1: Mean native beta reward component over 60 steps, all phase offsets retained. Each entry is one actual deterministic simulator run for its seed and phase. These are not independent training replicates or a confidence interval.

Environment seed	phase 0	phase 1	phase 2	phase 3
1331	16.0806	16.5743	16.3820	16.0847
1447	18.0763	18.1054	17.9033	17.9740

Under the wrapper’s decomposition, the native reward is the negative beta component minus .05 times the binary ON action. Since all phases share ON fraction .5, their mean action penalty is .025 and all reward differences within a seed come from the recorded beta component. The four-phase beta range is .4936 for seed 1331 and .2021 for seed 1447. The minimum is phase 0 in the first seed but phase 2 in the second. A phase selected on one initial condition does not become a generally best waveform. Reporting only the best of the eight runs would hide the seed dependence and create selection bias.

All eight LFP traces have different saved SHA-256 hashes, consistent with a changed simulated trajectory rather than merely relabeling the same observation. Hash differences alone do not measure clinical effect or numerical error. The state equations, reward scale and sampling window remain the upstream implementation; a large raw beta component should not be compared to clinical symptom severity. At the time of this phase plan there were no matched OFF/ON controls; the separate frozen control audit below supplies them without changing or selecting these eight outcomes. The earlier frozen-policy cost-envelope negative remains untouched.

The plan is stored in `src/native_phase_plan.json`, each bounded run in `src/native_phase_audit.py`, and the complete merged grid in `results/native_phase_audit.json`. Individual files retain the actions and per-step rewards, source commit, plan hash, LFP hash, switch

count and duration. Tests verify all eight expected phase/seed pairs, exact action schedules, reward decomposition and deterministic aggregation. CPU JAX 0.5.3 and Diffrax 0.7.0 were installed for this rerun with the pinned source; that is environment preparation, not a new physiological validation. No prior model, weights or result file is replaced.

The narrow result is that phase alignment changes finite-horizon outcomes at equal duty and action-magnitude cost in this native simulator. The order of phases changes with seed. Longer horizons, repeated initialization distributions, numerical tolerances and matched new OFF/ON baselines would be needed for a stable simulator-policy conclusion. Published SAC protocol reproduction, independently validated neural dynamics and clinical treatment inference remain separate open gaps. This audit adds actual method and result analysis to the body; appendix tables and total PDF length are not counted toward the owner’s fifty-plus research-content-page gate.

42.1 Matched sixty-step controls retain the suppression–cost tradeoff

A separately frozen plan restarts the identical native environment at each of the two phase seeds and applies either sixty OFF or sixty ON actions. All four runs use the same sixty-step horizon, reward, source commit and initialization wrapper. They charge zero and 300 V-step action magnitude respectively. This matched comparison does not pool the older 150-step experiments. Every phase

Table 2: Matched native controls. Half-duty columns give the complete four-phase range, not a selected best policy. Lower beta component means greater suppression within this simulator.

Seed	all-OFF beta	half-duty beta range	all-ON beta
1331	19.0657	16.0806–16.5743	13.8673
1447	20.7061	17.9033–18.1054	16.2394

suppresses the recorded beta component relative to OFF, but no half-duty phase suppresses it as much as ON. All-ON also has the better native reward at this short horizon, despite its .05 mean action penalty. Half-duty costs 150 rather than 300 V-step units, so a lesser action budget accompanies lesser suppression. This is a tradeoff, not evidence that periodic stimulation wins under the native scalar reward or that a learned controller has improved. Selecting a different penalty after viewing these outcomes would change the objective after the test. The ratio of OFF-to-half-duty suppression to OFF-to-ON suppression is descriptive within each deterministic seed and phase. It does not identify a clinical dose response, confidence interval or efficient physical-energy frontier. Two seeds cannot establish stable ordering over initialization distributions. The earlier failed learned-policy evaluation and duty-envelope negative remain unchanged. The control plans, four complete reward/action ledgers and reproducible merger are saved in `src/native_matched_plan.json`, `src/native_matched_audit.py` and `results/native_matched_audit.json`. Tests recheck exact action counts, reward decomposition, source/plan identity and complete outcome retention.

42.2 Four fresh seeds break a universal short-horizon ordering

A further frozen plan adds four fresh environment seeds, each restarted for OFF, ON and all four half-duty phases. All 24 actual native runs retain the same sixty-step horizon, source, solver and

reward. No phase is selected for training or deployment. This grid is a descriptive initialization sensitivity check, not a confidence interval, numerical-tolerance audit or clinical validation. The earlier

Table 3: All six policies at each of four new native seeds. Beta component over sixty steps; OFF/half-duty/ON costs remain 0/150/300 V-step.

Seed	OFF	ON	phase 0	phase 1	phase 2	phase 3
1561	26.6949	26.3673	26.8264	26.5430	26.5538	26.7620
1687	16.1532	13.7162	15.0994	15.4641	15.3645	15.1431
1801	38.9012	28.1422	31.9206	32.3874	32.3568	32.0978
1931	16.4914	16.1929	15.4298	15.4225	15.4348	15.4257

matched result that every half-duty phase lies between OFF and ON holds for seeds 1331 and 1447 only. Seed 1561 has phases zero and three worse than OFF despite nonzero stimulation cost. Seed 1931 reverses the half-duty-versus-ON ordering: all four phases have lower beta component and better native reward than ON at half its action-magnitude cost. Seeds 1687 and 1801 retain the intermediate half-duty ordering. Hence neither universal superiority nor universal inferiority of half-duty follows from this simulator. A fixed periodic waveform can interact differently with different initialized native dynamics. The eight-phase suppression ratio from the previous two-seed audit is not extended into a population effect. Ratios become unstable when OFF and ON are close, as in the new 1561 and 1931 outcomes. A selected winning seed or phase would conceal reversals. All 24 complete action/reward files and LFP hashes remain in `results/native_broad_*`; the frozen plan and deterministic merger are `src/native_broad_plan.json` and `src/merge_native_broad.py`. Tests require every seed/policy pair, exact schedules, costs, reward decomposition and the observed negative/reversal cases. No learned controller or prior weights change. Longer windows, broader initialization distributions and numerical/model checks are needed before a stable policy conclusion.

42.3 A selected-seed horizon extension tests persistence, not independent replication

After observing the sixty-step reversals, we deliberately select seeds 1561 and 1931 for a separately frozen 120-step extension. This selection makes the experiment exploratory, not a fresh held-out confirmation. OFF, ON and phases zero and one are fixed before the new outcomes; phases two and three are not run at 120 steps. Eight actual native runs restart the same environment and preserve every selected outcome. Costs now equal 0, 300 and 600 V-step for OFF, half-duty and ON, so absolute action totals cannot be silently pooled with sixty-step totals. Seed 1561 no longer has phase zero

Table 4: Mean native beta component over 120 steps at two seeds selected because of the prior reversals. These are conditional follow-ups, not a population estimate.

Selected seed	OFF	ON	phase 0	phase 1
1561	21.5398	18.7820	20.9909	20.3160
1931	14.5805	13.1297	12.2113	12.1780

worse than OFF over this doubled averaging window: both retained half-duty phases lie between OFF and ON. Thus its sixty-step negative is horizon-dependent. At seed 1931 both retained phases remain below ON in beta component and have better native reward at half its action-magnitude cost. This persistence over one longer window does not establish stationary benefit, an optimal waveform, or a learned-controller improvement. The averaged transient trajectory and initialization remain part of the estimand. Plans, source identity, actions, rewards and LFP hashes are retained in `src/native_horizon_plan.json` and `results/native_horizon_*`; tests reproduce complete aggregation, 120-step schedules, cost and reward decomposition. The choice of seeds and omitted phases is explicit rather than hidden as favorable post-hoc confirmation. Larger predeclared horizons, independent seed distributions, solver/model checks and real physiological validation remain needed. All earlier failed learned-policy results and phase reversals remain visible.

A Numbered formula compendium

Every quantitative claim in the body traces to one of the following definitions. Numbering continues from the body.

A.1 Fitted-Q control in the surrogate

The surrogate return over horizon T with per-step energy and burst penalties λ_e, λ_b is

$$G = \sum_{t=0}^{T-1} r_t, \quad r_t = -(\beta_t^2 + \lambda_e a_t^2 + \lambda_b b_t), \quad (15)$$

with β_t the simulated beta-band amplitude, a_t the stimulation amplitude and b_t the burst indicator. The Bellman optimality equation backing the fitted-Q update is

$$Q^*(s, a) = \mathbb{E} \left[r + \gamma \max_{a'} Q^*(s', a') \right], \quad (16)$$

and the fitted-Q regression minimizes the TD error over the replay batch $\mathcal{B}$:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{B}|} \sum_{(s,a,r,s') \in \mathcal{B}} \left(r + \gamma \max_{a'} Q_{\bar{\theta}}(s', a') - Q_{\theta}(s, a) \right)^2. \quad (17)$$

A.2 Policy-gradient pilot

The native REINFORCE pilot maximizes

$$\nabla_{\theta} J = \mathbb{E} \left[\sum_t \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (G_t - b_t) \right], \quad (18)$$

with baseline b_t the running return mean; the Adam variant uses moment estimates $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$.

A.3 Tremor-trace descriptors

TremorDB traces are summarized by RMS and 4–6 Hz Welch power:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}, \quad P_{4-6} = \frac{1}{K} \sum_k \frac{1}{NU} \left| \sum_n w_n x_n e^{-j2\pi f_k n / f_s} \right|^2. \quad (19)$$

A.4 Expression-audit statistics

Per-probe differential signal uses Welch’s t :

$$t_p = \frac{\bar{x}_p^{PD} - \bar{x}_p^{CTL}}{\sqrt{s_{PD,p}^2/n_{PD} + s_{CTL,p}^2/n_{CTL}}}. \quad (20)$$

Held-out discrimination uses the rank-form AUC:

$$\text{AUC} = \frac{1}{n_+ n_-} \sum_{i: +} \sum_{j: -} \mathbb{I}[\hat{s}_i > \hat{s}_j] + \frac{1}{2} \mathbb{I}[\hat{s}_i = \hat{s}_j]. \quad (21)$$

Label-permutation nulls over B permutations give the exceedance fraction

$$\hat{p} = \frac{1 + \sum_b \mathbb{I}[\text{AUC}_b \geq \text{AUC}_{real}]}{B + 1}. \quad (22)$$

Signature overlap uses the Jaccard index

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (23)$$

B Distillation-law runs (complete, 240 runs)

Every run behind the distillation-law margin analysis. Machine-readable record: `results/sepsis_distill_law.json`.

Table 5: Distillation-law runs, all arms.

Arm	Width	Epochs	Seed	Return	Agreement	End loss
plain	16	50	0	0.7830	0.0461	0.0004
plain	16	50	1	0.7850	0.0838	0.0006
plain	16	50	3	0.7866	0.0670	0.0006
plain	16	50	2	0.7843	0.0391	0.0004
plain	16	50	4	0.7809	0.0545	0.0004
plain	16	50	5	0.7864	0.0740	0.0005
plain	16	50	6	0.7873	0.0754	0.0006
plain	16	50	7	0.7825	0.0321	0.0005

continued

Arm	Width	Epochs	Seed	Return	Agreement	End loss
plain	16	50	8	0.7830	0.0447	0.0005
plain	16	50	9	0.7820	0.0391	0.0004
plain	16	150	0	0.8023	0.1494	0.0002
plain	16	150	1	0.8084	0.1690	0.0002
plain	16	150	2	0.7944	0.0754	0.0002
plain	16	150	3	0.8027	0.1439	0.0002
plain	16	150	4	0.7969	0.1131	0.0002
plain	16	150	5	0.8008	0.1494	0.0002
plain	16	150	6	0.8109	0.2039	0.0002
plain	16	150	7	0.7935	0.0950	0.0002
plain	16	150	8	0.8006	0.1006	0.0001
plain	16	150	9	0.7954	0.1145	0.0002
plain	16	500	0	0.8218	0.2835	4.02e-05
plain	16	500	1	0.8209	0.2612	5.33e-05
plain	16	500	2	0.8215	0.2765	3.53e-05
plain	16	500	3	0.8245	0.2528	6.07e-05
plain	16	500	4	0.8300	0.3003	3.26e-05
plain	16	500	5	0.8230	0.2584	3.90e-05
plain	16	500	6	0.8261	0.2765	4.04e-05
plain	16	500	7	0.8221	0.2612	4.12e-05
plain	16	500	8	0.8278	0.2779	3.54e-05
plain	16	500	9	0.8235	0.2877	3.89e-05
plain	64	50	0	0.7848	0.0489	0.0004
plain	64	50	1	0.7891	0.0684	0.0004
plain	64	50	2	0.7855	0.0587	0.0004
plain	64	50	3	0.7855	0.0726	0.0004
plain	64	50	4	0.7803	0.0405	0.0003
plain	64	50	5	0.7840	0.0405	0.0004
plain	64	50	6	0.7850	0.0559	0.0004
plain	64	50	7	0.7851	0.0601	0.0004
plain	64	50	8	0.7846	0.0461	0.0004
plain	64	50	9	0.7865	0.0628	0.0004
plain	64	150	0	0.8105	0.1690	0.0001
plain	64	150	1	0.8101	0.1620	0.0001
plain	64	150	2	0.8103	0.1550	8.76e-05
plain	64	150	3	0.8133	0.1606	0.0001
plain	64	150	4	0.8064	0.1369	4.37e-05
plain	64	150	5	0.8104	0.1453	0.0001
plain	64	150	6	0.8086	0.1425	8.70e-05
plain	64	150	7	0.8109	0.1592	8.25e-05
plain	64	150	8	0.8085	0.1397	0.0001
plain	64	150	9	0.8019	0.1327	0.0001
plain	64	500	0	0.8573	0.4246	1.23e-05
plain	64	500	1	0.8580	0.4385	1.13e-05
plain	64	500	2	0.8583	0.4581	3.83e-05
plain	64	500	3	0.8590	0.4539	3.21e-05
plain	64	500	4	0.8600	0.4707	6.11e-06
plain	64	500	6	0.8580	0.4232	8.63e-05
plain	64	500	5	0.8576	0.4232	1.69e-05
plain	64	500	7	0.8574	0.4623	2.28e-05
plain	64	500	8	0.8569	0.4302	9.50e-06
plain	128	50	0	0.7892	0.0545	0.0003

continued

Arm	Width	Epochs	Seed	Return	Agreement	End loss
plain	128	50	1	0.7885	0.0573	0.0003
plain	64	500	9	0.8562	0.4008	3.57e-05
plain	128	50	2	0.7885	0.0754	0.0003
plain	128	50	3	0.7866	0.0628	0.0003
plain	128	50	4	0.7903	0.0810	0.0002
plain	128	50	5	0.7873	0.0642	0.0003
plain	128	50	6	0.7921	0.0740	0.0003
plain	128	50	7	0.7872	0.0656	0.0003
plain	128	50	8	0.7888	0.0670	0.0003
plain	128	50	9	0.7849	0.0628	0.0003
plain	128	150	0	0.8275	0.2612	2.80e-05
plain	128	150	1	0.8297	0.2514	6.00e-05
plain	128	150	2	0.8322	0.2598	5.98e-05
plain	128	150	3	0.8277	0.2528	7.17e-05
plain	128	150	4	0.8396	0.3017	4.03e-05
plain	128	150	5	0.8323	0.2696	6.25e-05
plain	128	150	6	0.8317	0.2612	5.42e-05
plain	128	150	7	0.8294	0.2514	2.87e-05
plain	128	150	8	0.8341	0.2877	6.43e-05
plain	128	150	9	0.8268	0.2528	6.54e-05
plain	128	500	0	0.8730	0.5866	1.33e-06
plain	128	500	1	0.8713	0.5642	1.63e-05
plain	128	500	2	0.8676	0.5084	8.51e-06
plain	128	500	3	0.8651	0.4693	0.0002
plain	128	500	4	0.8731	0.5922	2.70e-05
plain	128	500	5	0.8701	0.5670	3.19e-05
plain	128	500	7	0.8717	0.5615	1.71e-05
plain	128	500	6	0.8708	0.5768	1.92e-05
plain	128	500	8	0.8710	0.5726	6.11e-05
plain	128	500	9	0.8693	0.5391	2.17e-05
plain	256	50	0	0.7953	0.0964	0.0002
plain	256	50	1	0.7928	0.0894	0.0002
plain	256	50	2	0.7962	0.0866	0.0002
plain	256	50	3	0.7974	0.0908	0.0002
plain	256	50	4	0.7985	0.0782	0.0002
plain	256	50	5	0.7926	0.0712	0.0002
plain	256	50	6	0.7959	0.0964	0.0002
plain	256	50	7	0.7957	0.0880	0.0002
plain	256	50	8	0.7973	0.0908	0.0002
plain	256	50	9	0.7972	0.1034	0.0002
plain	256	150	0	0.8604	0.4511	2.26e-05
plain	256	150	1	0.8582	0.4274	2.54e-05
plain	256	150	2	0.8627	0.4679	2.44e-05
plain	256	150	3	0.8624	0.4595	2.09e-05
plain	256	150	4	0.8592	0.4372	2.25e-05
plain	256	150	5	0.8577	0.4204	2.37e-05
plain	256	150	6	0.8610	0.4609	2.25e-05
plain	256	150	7	0.8612	0.4609	2.39e-05
plain	256	150	8	0.8590	0.4330	2.42e-05
plain	256	150	9	0.8626	0.4735	2.13e-05
plain	256	500	0	0.8668	0.5223	0.0001
plain	256	500	1	0.8700	0.5489	9.43e-05

continued

Arm	Width	Epochs	Seed	Return	Agreement	End loss
plain	256	500	2	0.8707	0.5656	0.0001
plain	256	500	3	0.8708	0.5475	0.0002
plain	256	500	5	0.8734	0.6061	5.53e-05
plain	256	500	4	0.8741	0.6047	3.91e-05
plain	256	500	6	0.8737	0.6089	2.51e-05
plain	256	500	7	0.8730	0.5894	0.0001
plain	256	500	8	0.8741	0.5992	2.55e-05
plain	256	500	9	0.8734	0.6187	4.09e-05
margin_weighted	16	50	0	0.7830	0.0461	0.0004
margin_weighted	16	50	1	0.7848	0.0838	0.0006
margin_weighted	16	50	2	0.7844	0.0405	0.0004
margin_weighted	16	50	3	0.7859	0.0642	0.0006
margin_weighted	16	50	4	0.7805	0.0503	0.0004
margin_weighted	16	50	5	0.7869	0.0740	0.0005
margin_weighted	16	50	6	0.7872	0.0740	0.0006
margin_weighted	16	50	7	0.7829	0.0321	0.0005
margin_weighted	16	50	8	0.7824	0.0433	0.0005
margin_weighted	16	50	9	0.7822	0.0391	0.0004
margin_weighted	16	150	0	0.8020	0.1480	0.0001
margin_weighted	16	150	1	0.8085	0.1648	0.0002
margin_weighted	16	150	2	0.7948	0.0782	0.0002
margin_weighted	16	150	3	0.8023	0.1453	0.0002
margin_weighted	16	150	4	0.7972	0.1131	0.0002
margin_weighted	16	150	5	0.8000	0.1453	0.0002
margin_weighted	16	150	6	0.8103	0.1997	0.0002
margin_weighted	16	150	7	0.7942	0.0922	0.0002
margin_weighted	16	150	8	0.8011	0.1034	0.0001
margin_weighted	16	150	9	0.7960	0.1131	0.0002
margin_weighted	16	500	0	0.8238	0.2849	4.05e-05
margin_weighted	16	500	1	0.8289	0.2933	2.99e-05
margin_weighted	16	500	2	0.8218	0.2779	3.49e-05
margin_weighted	16	500	3	0.8224	0.2374	6.14e-05
margin_weighted	16	500	4	0.8222	0.2444	3.65e-05
margin_weighted	16	500	5	0.8209	0.2486	4.06e-05
margin_weighted	16	500	6	0.8253	0.2682	4.09e-05
margin_weighted	16	500	7	0.8216	0.2668	4.38e-05
margin_weighted	16	500	8	0.8297	0.2947	3.42e-05
margin_weighted	16	500	9	0.8302	0.3128	3.77e-05
margin_weighted	64	50	0	0.7851	0.0489	0.0004
margin_weighted	64	50	1	0.7889	0.0670	0.0004
margin_weighted	64	50	2	0.7851	0.0601	0.0004
margin_weighted	64	50	3	0.7860	0.0726	0.0004
margin_weighted	64	50	4	0.7804	0.0419	0.0004
margin_weighted	64	50	5	0.7842	0.0405	0.0004
margin_weighted	64	50	6	0.7855	0.0587	0.0004
margin_weighted	64	50	7	0.7855	0.0587	0.0004
margin_weighted	64	50	8	0.7846	0.0461	0.0004
margin_weighted	64	50	9	0.7858	0.0656	0.0005
margin_weighted	64	150	0	0.8098	0.1662	0.0001
margin_weighted	64	150	1	0.8093	0.1606	0.0001
margin_weighted	64	150	2	0.8096	0.1522	8.70e-05
margin_weighted	64	150	3	0.8125	0.1620	0.0001

continued

Arm	Width	Epochs	Seed	Return	Agreement	End loss
margin_weighted	64	150	4	0.8050	0.1355	4.39e-05
margin_weighted	64	150	5	0.8095	0.1425	0.0001
margin_weighted	64	150	6	0.8088	0.1397	8.65e-05
margin_weighted	64	150	7	0.8096	0.1536	8.26e-05
margin_weighted	64	150	8	0.8086	0.1439	0.0001
margin_weighted	64	150	9	0.8031	0.1355	0.0001
margin_weighted	64	500	0	0.8560	0.4204	1.62e-05
margin_weighted	64	500	1	0.8572	0.4204	5.45e-05
margin_weighted	64	500	2	0.8567	0.4399	5.82e-05
margin_weighted	64	500	3	0.8589	0.4483	3.45e-05
margin_weighted	64	500	4	0.8598	0.4665	6.88e-06
margin_weighted	64	500	5	0.8553	0.4008	5.56e-05
margin_weighted	64	500	6	0.8596	0.4525	5.81e-05
margin_weighted	64	500	7	0.8567	0.4595	4.09e-05
margin_weighted	64	500	8	0.8570	0.4316	7.84e-06
margin_weighted	64	500	9	0.8555	0.3980	4.57e-05
margin_weighted	128	50	0	0.7895	0.0559	0.0003
margin_weighted	128	50	1	0.7889	0.0587	0.0003
margin_weighted	128	50	2	0.7871	0.0656	0.0003
margin_weighted	128	50	3	0.7867	0.0615	0.0003
margin_weighted	128	50	4	0.7907	0.0810	0.0003
margin_weighted	128	50	5	0.7871	0.0628	0.0003
margin_weighted	128	50	6	0.7921	0.0712	0.0003
margin_weighted	128	50	7	0.7882	0.0670	0.0003
margin_weighted	128	50	9	0.7850	0.0601	0.0003
margin_weighted	128	50	8	0.7894	0.0642	0.0003
margin_weighted	128	150	1	0.8309	0.2598	5.96e-05
margin_weighted	128	150	0	0.8269	0.2584	2.79e-05
margin_weighted	128	150	2	0.8301	0.2472	5.93e-05
margin_weighted	128	150	3	0.8272	0.2514	7.08e-05
margin_weighted	128	150	4	0.8394	0.3003	4.98e-05
margin_weighted	128	150	5	0.8304	0.2542	6.48e-05
margin_weighted	128	150	6	0.8307	0.2556	5.29e-05
margin_weighted	128	150	7	0.8292	0.2458	2.81e-05
margin_weighted	128	150	8	0.8343	0.2863	6.37e-05
margin_weighted	128	150	9	0.8277	0.2528	6.56e-05
margin_weighted	128	500	0	0.8727	0.5852	1.44e-06
margin_weighted	128	500	1	0.8699	0.5447	2.27e-05
margin_weighted	128	500	2	0.8669	0.5000	6.89e-05
margin_weighted	128	500	3	0.8640	0.4679	0.0002
margin_weighted	128	500	4	0.8653	0.4972	0.0001
margin_weighted	128	500	5	0.8697	0.5545	2.24e-05
margin_weighted	128	500	6	0.8708	0.5824	5.53e-05
margin_weighted	128	500	7	0.8720	0.5754	1.01e-05
margin_weighted	128	500	8	0.8693	0.5559	0.0001
margin_weighted	128	500	9	0.8681	0.5293	2.95e-05
margin_weighted	256	50	0	0.7945	0.0908	0.0002
margin_weighted	256	50	1	0.7913	0.0852	0.0002
margin_weighted	256	50	3	0.7977	0.0922	0.0002
margin_weighted	256	50	2	0.7954	0.0838	0.0002
margin_weighted	256	50	4	0.7987	0.0810	0.0002
margin_weighted	256	50	5	0.7927	0.0712	0.0002

continued

Arm	Width	Epochs	Seed	Return	Agreement	End loss
margin_weighted	256	50	6	0.7964	0.0964	0.0002
margin_weighted	256	50	7	0.7952	0.0866	0.0002
margin_weighted	256	50	8	0.7971	0.0894	0.0002
margin_weighted	256	50	9	0.7974	0.1061	0.0002
margin_weighted	256	150	0	0.8604	0.4539	2.23e-05
margin_weighted	256	150	1	0.8581	0.4218	2.54e-05
margin_weighted	256	150	2	0.8625	0.4665	2.42e-05
margin_weighted	256	150	3	0.8629	0.4525	2.06e-05
margin_weighted	256	150	5	0.8575	0.4190	2.34e-05
margin_weighted	256	150	4	0.8592	0.4344	2.23e-05
margin_weighted	256	150	7	0.8588	0.4497	2.38e-05
margin_weighted	256	150	6	0.8611	0.4609	2.23e-05
margin_weighted	256	150	9	0.8624	0.4707	2.14e-05
margin_weighted	256	150	8	0.8592	0.4344	2.39e-05
margin_weighted	256	500	1	0.8737	0.6006	6.69e-05
margin_weighted	256	500	0	0.8735	0.6075	1.62e-05
margin_weighted	256	500	2	0.8679	0.5265	7.63e-05
margin_weighted	256	500	3	0.8742	0.6215	4.91e-05
margin_weighted	256	500	5	0.8722	0.5838	9.53e-05
margin_weighted	256	500	4	0.8730	0.5964	5.60e-05
margin_weighted	256	500	6	0.8724	0.5838	3.36e-05
margin_weighted	256	500	7	0.8741	0.6047	5.29e-05
margin_weighted	256	500	8	0.8742	0.6089	7.33e-05
margin_weighted	256	500	9	0.8746	0.6383	1.25e-05

C TremorDB trace records (complete, 55 records)

Public fixed-condition parkinsonian tremor traces; instrument units, no adaptive-policy data.
Machine-readable record:

results/parkinson_tremordb.json.

Table 6: TremorDB records.

File	Group	Subject	DBS	Med	n	RMS
g1ren.let	renh	g1	on	on	6234	4.595
g2ren.rit	renh	g2	on	on	7680	5.419
s6ren.let	renh	s6	on	on	6288	3.776
s7ren.rit	renh	s7	on	on	6468	3.203
s8ren.let	renh	s8	on	on	6252	3.486
g9ren.rit	renl	g9	on	on	6276	10.088
g10ren.let	renl	g10	on	on	6180	11.443
g11ren.let	renl	g11	on	on	6276	2.699
g12ren.rit	renl	g12	on	on	6492	3.095
g13ren.rit	renl	g13	on	on	6282	4.103
s14ren.rit	renl	s14	on	on	6408	3.040
s15ren.rit	renl	s15	on	on	7284	2.410
s16ren.let	renl	s16	on	on	6300	2.671
g1ref.let	refh	g1	on	off	6690	17.533
g2ref.rit	refh	g2	on	off	6336	5.112
s6ref.let	refh	s6	on	off	7260	19.668

continued

File	Group	Subject	DBS	Med	n	RMS
s7ref.rit	refh	s7	on	off	6402	14.313
s8ref.let	refh	s8	on	off	7896	7.397
g9ref.rit	refl	g9	on	off	6174	12.520
g10ref.let	refl	g10	on	off	6252	4.131
g11ref.let	refl	g11	on	off	6396	3.504
g12ref.rit	refl	g12	on	off	6714	2.639
g13ref.rit	refl	g13	on	off	6564	3.332
s14ref.rit	refl	s14	on	off	6282	7.664
s15ref.rit	refl	s15	on	off	6228	2.398
s16ref.let	refl	s16	on	off	7128	6.234
g1ron.let	ronh	g1	off	on	6264	3.535
g2ron.rit	ronh	g2	off	on	7134	9.582
v4ron.rit	ronh	v4	off	on	6432	0.206
v5ron.let	ronh	v5	off	on	7344	0.048
s6ron.let	ronh	s6	off	on	7356	13.394
s7ron.rit	ronh	s7	off	on	6234	3.463
s8ron.let	ronh	s8	off	on	6720	2.657
g9ron.rit	ronl	g9	off	on	9156	4.535
g10ron.let	ronl	g10	off	on	12108	20.654
g11ron.let	ronl	g11	off	on	6366	3.996
g12ron.rit	ronl	g12	off	on	7176	2.995
g13ron.rit	ronl	g13	off	on	6234	4.184
s14ron.rit	ronl	s14	off	on	6228	2.688
s15ron.rit	ronl	s15	off	on	6102	3.358
s16ron.let	ronl	s16	off	on	6696	3.338
g2rof.rit	rof	g2	off	off	7176	0.312
v4rof.rit	rof	v4	off	off	6282	0.178
v5rof.let	rof	v5	off	off	3400	0.088
s6rof.let	rof	s6	off	off	9114	0.274
s7rof.rit	rof	s7	off	off	6264	52.931
s8rof.let	rof	s8	off	off	6216	0.131
g9rof.rit	rof	g9	off	off	6354	5.433
g10rof.let	rof	g10	off	off	6180	4.582
g11rof.let	rof	g11	off	off	6306	6.665
g12rof.rit	rof	g12	off	off	6360	3.104
g13rof.rit	rof	g13	off	off	7230	6.067
s14rof.rit	rof	s14	off	off	6492	3.419
s15rof.rit	rof	s15	off	off	6300	2.427
s16rof.let	rof	s16	off	off	6594	5.640

D Per-probe blood-expression audit (top 300 by $|t|$, train split)

Welch t contrasting PD and control whole-blood expression over the pooled 438-sample GEO matrix (train split only; GSE99039/6613/72267/22491). Full matrix reproducible from `scripts/` plus the GEO series matrices. Train split: 140 PD, 153 control, 14899 probes.

Table 7: Top 1200 probes by $|t|$ on the train split.

Probe	mean PD	mean CTL	Welch t	$ t $ rank
209841_s.at	4.917	5.520	-5.29	1
209840_s.at	5.808	6.490	-5.01	2
208932.at	8.907	8.699	4.46	3
204576_s.at	5.839	5.987	-4.26	4
240737.at	6.460	6.278	4.25	5
217929_s.at	7.009	6.811	4.23	6
212187_x.at	6.924	6.609	4.05	7
207384.at	7.952	7.634	4.01	8
216388_s.at	5.939	5.740	3.92	9
205936_s.at	8.064	7.858	3.88	10
218130.at	8.677	8.510	3.86	11
210890_x.at	6.544	6.354	3.74	12
210136.at	9.980	9.673	3.74	13
90265.at	9.909	9.801	3.69	14
225251.at	9.086	8.888	3.67	15
1554544_a.at	9.354	9.054	3.66	16
1555736_a.at	9.462	9.264	3.64	17
203936_s.at	9.669	9.295	3.63	18
203712.at	5.792	5.924	-3.62	19
205367.at	7.643	7.429	3.62	20
210401.at	7.804	7.586	3.61	21
203413.at	7.576	7.863	-3.61	22
213988_s.at	10.136	9.970	3.58	23
238053.at	5.666	5.457	3.58	24
212443.at	8.375	8.140	3.57	25
218638_s.at	7.933	7.650	3.57	26
225059.at	8.908	8.746	3.55	27
211748_x.at	8.082	7.831	3.55	28
203028_s.at	10.982	10.823	3.55	29
211178_s.at	8.219	8.057	3.53	30
210423_s.at	9.601	9.385	3.52	31
206208.at	6.176	5.979	3.52	32
242167.at	6.567	6.408	3.50	33
215639.at	6.257	5.984	3.49	34
202671_s.at	7.282	7.132	3.47	35
204158_s.at	9.389	9.159	3.45	36
207536_s.at	5.875	5.671	3.43	37
207697_x.at	9.098	8.933	3.42	38
204436.at	8.796	8.652	3.42	39
211543_s.at	7.928	7.771	3.39	40
201053_s.at	7.920	8.114	-3.39	41
1558426_x.at	9.074	8.907	3.38	42
218511_s.at	6.283	6.374	-3.38	43
229029.at	5.761	6.028	-3.34	44
210140.at	9.876	9.646	3.33	45
214084_x.at	10.544	10.390	3.31	46
239893.at	8.474	8.244	3.31	47
220751_s.at	7.785	8.063	-3.29	48
218510_x.at	5.755	5.926	-3.29	49
229389.at	10.054	9.672	3.29	50

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
204246_s.at	8.407	8.298	3.29	51
209929_s.at	8.000	7.887	3.28	52
236292.at	6.954	6.744	3.28	53
1552667_a.at	7.150	6.935	3.27	54
221815.at	7.117	6.875	3.26	55
211687_x.at	5.822	5.655	3.26	56
203760_s.at	10.187	9.980	3.26	57
204039.at	7.232	7.063	3.25	58
212821.at	6.345	6.156	3.25	59
216112.at	5.953	5.847	3.25	60
211955.at	6.572	6.732	-3.25	61
234672_s.at	5.612	5.767	-3.25	62
45687.at	8.630	8.522	3.24	63
217611.at	6.256	6.028	3.24	64
211133_x.at	9.105	8.939	3.24	65
219680.at	7.762	7.654	3.23	66
228008.at	7.057	6.900	3.23	67
220001.at	8.605	8.384	3.22	68
202574_s.at	8.486	8.338	3.21	69
224130_s.at	7.970	7.857	3.21	70
235816_s.at	6.390	6.168	3.21	71
223482.at	8.026	7.858	3.21	72
201379_s.at	8.878	8.735	3.21	73
227626.at	7.946	8.124	-3.21	74
1557675.at	8.082	7.882	3.20	75
222199_s.at	7.677	7.557	3.20	76
204040.at	5.516	5.700	-3.20	77
202025_x.at	7.490	7.407	3.20	78
202096_s.at	10.187	10.037	3.19	79
204961_s.at	10.914	10.752	3.17	80
1563071.at	5.864	5.709	3.17	81
203317.at	7.846	7.709	3.17	82
210784_x.at	9.894	9.729	3.17	83
206209_s.at	6.637	6.392	3.16	84
204951.at	7.103	7.270	-3.16	85
223448_x.at	5.670	5.793	-3.16	86
204193.at	7.817	7.650	3.16	87
242714.at	5.725	5.512	3.15	88
218388.at	9.335	9.226	3.15	89
217796_s.at	8.156	8.054	3.15	90
223562.at	9.016	8.882	3.14	91
201702_s.at	6.466	6.294	3.14	92
227833_s.at	9.317	9.157	3.14	93
220000.at	8.844	8.652	3.14	94
225637.at	9.856	9.719	3.13	95
229373.at	8.261	8.057	3.13	96
206722_s.at	7.165	7.010	3.13	97
207104_x.at	7.561	7.404	3.13	98
1555311.at	7.371	7.271	3.12	99
218206_x.at	7.689	7.538	3.12	100
204122.at	11.668	11.517	3.11	101
231830_x.at	8.205	8.042	3.11	102

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
211135_x.at	9.887	9.727	3.11	103
217497.at	7.955	7.618	3.11	104
210580_x.at	8.557	8.422	3.10	105
209316_s.at	6.016	6.164	-3.10	106
237561_x.at	6.159	6.022	3.10	107
1555088_x.at	7.569	7.416	3.09	108
240780.at	5.975	5.868	3.09	109
225955.at	6.759	6.575	3.09	110
203386.at	5.677	5.978	-3.09	111
222581.at	5.860	5.971	-3.09	112
206324_s.at	6.624	6.493	3.09	113
210128_s.at	6.185	6.054	3.09	114
212115.at	7.161	7.262	-3.09	115
242040.at	6.673	6.513	3.08	116
214131.at	4.412	3.757	3.07	117
204929_s.at	6.547	6.398	3.07	118
213045.at	8.009	7.848	3.07	119
203685.at	6.615	6.867	-3.06	120
205418.at	7.230	7.062	3.06	121
212576.at	8.427	8.283	3.05	122
224588.at	7.274	8.577	-3.05	123
221250_s.at	6.913	6.811	3.05	124
225602.at	9.512	9.403	3.04	125
1562289.at	7.629	7.405	3.04	126
218831_s.at	9.341	9.164	3.04	127
1567628.at	6.488	6.290	3.04	128
1553723.at	8.121	7.825	3.03	129
227615.at	6.333	6.500	-3.03	130
57082.at	8.739	8.861	-3.02	131
1554443_s.at	5.999	5.858	3.02	132
225503.at	7.582	7.480	3.02	133
217147_s.at	5.646	6.002	-3.02	134
226272.at	7.804	8.133	-3.02	135
212252.at	7.077	6.899	3.01	136
218660.at	9.645	9.448	3.01	137
225043.at	8.926	8.775	3.01	138
205241.at	7.863	7.646	3.01	139
1555086.at	7.361	7.181	3.01	140
201400.at	10.352	10.213	3.01	141
201393_s.at	9.550	9.363	3.00	142
227811.at	10.666	10.547	3.00	143
211951.at	6.984	7.164	-3.00	144
203402.at	8.059	7.939	3.00	145
212860.at	9.416	9.276	3.00	146
211339_s.at	8.460	8.734	-3.00	147
203669_s.at	6.916	6.740	3.00	148
239333_x.at	6.136	6.000	3.00	149
212457.at	9.008	8.854	2.99	150
215299_x.at	9.676	9.492	2.99	151
234326.at	6.940	6.689	2.99	152
201995.at	5.802	5.669	2.98	153
211397_x.at	6.848	6.666	2.98	154

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
229871_at	7.957	7.786	2.98	155
211336_x_at	7.473	7.329	2.98	156
217850_at	5.573	5.791	-2.98	157
236846_at	6.049	5.805	2.97	158
201139_s_at	6.161	6.402	-2.97	159
212284_x_at	12.781	12.964	-2.96	160
221807_s_at	6.361	6.228	2.96	161
232420_x_at	6.805	6.669	2.96	162
204858_s_at	7.891	7.710	2.96	163
207122_x_at	8.039	7.908	2.96	164
203459_s_at	7.331	7.232	2.95	165
214907_at	6.934	6.750	2.95	166
242425_at	6.055	5.916	2.95	167
205147_x_at	10.119	9.963	2.95	168
1570410_at	6.149	6.039	2.95	169
228758_at	10.644	10.452	2.95	170
225273_at	8.989	8.812	2.95	171
211413_s_at	6.366	6.122	2.95	172
230528_s_at	6.686	6.487	2.95	173
210422_x_at	6.872	6.738	2.94	174
213187_x_at	12.704	12.634	2.94	175
212018_s_at	6.747	6.930	-2.94	176
210225_x_at	9.871	9.717	2.94	177
216210_x_at	9.007	8.910	2.94	178
205312_at	8.488	8.251	2.93	179
236280_at	5.996	6.365	-2.93	180
223944_at	8.324	8.173	2.93	181
1552691_at	7.947	7.766	2.93	182
226480_at	6.243	6.484	-2.93	183
227671_at	7.184	8.468	-2.93	184
226378_s_at	6.225	6.087	2.93	185
202659_at	9.322	9.191	2.92	186
212400_at	7.800	7.949	-2.92	187
218231_at	10.104	9.945	2.92	188
243051_at	5.956	5.788	2.92	189
212625_at	7.567	7.447	2.91	190
203514_at	9.361	9.179	2.91	191
228521_s_at	7.873	7.726	2.91	192
232190_x_at	6.400	6.280	2.91	193
243178_at	7.924	7.705	2.90	194
238715_at	6.037	5.899	2.90	195
224590_at	5.506	6.490	-2.90	196
218154_at	8.038	7.944	2.90	197
1562511_at	6.438	6.279	2.90	198
243495_s_at	6.932	6.468	2.90	199
201526_at	8.631	8.547	2.89	200
221728_x_at	5.937	6.754	-2.89	201
229841_at	6.586	6.407	2.89	202
204669_s_at	10.463	10.141	2.89	203
203370_s_at	7.257	7.028	2.88	204
221890_at	6.862	6.773	2.88	205
1557965_at	5.900	6.097	-2.88	206

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
202509_s.at	6.704	6.498	2.87	207
209137_s.at	8.606	8.427	2.87	208
226554.at	7.972	7.841	2.86	209
218920.at	7.906	7.775	2.86	210
206219_s.at	8.222	8.116	2.86	211
48031_r.at	6.718	6.884	-2.86	212
229726.at	6.197	6.296	-2.85	213
220112.at	5.615	5.811	-2.85	214
219577_s.at	7.154	6.980	2.85	215
202081.at	10.732	10.630	2.85	216
202283.at	5.775	5.910	-2.85	217
235309.at	5.764	5.881	-2.85	218
210706_s.at	9.726	9.439	2.84	219
201220_x.at	9.996	9.891	2.84	220
204650_s.at	6.165	6.061	2.84	221
220371_s.at	7.817	7.717	2.84	222
223445.at	6.816	6.719	2.84	223
225899_x.at	10.501	10.313	2.84	224
220712.at	6.892	6.722	2.84	225
227819.at	5.946	5.806	2.84	226
206545.at	5.780	6.065	-2.83	227
211250_s.at	7.274	7.137	2.83	228
222876_s.at	6.334	6.220	2.83	229
205774.at	5.784	5.674	2.83	230
226801_s.at	7.974	8.211	-2.83	231
224681.at	7.149	7.306	-2.83	232
239845.at	6.478	6.297	2.83	233
200827.at	7.174	7.039	2.83	234
238042.at	7.322	7.194	2.82	235
227533.at	5.667	5.896	-2.82	236
243182.at	7.166	6.836	2.82	237
227851_s.at	7.845	7.633	2.82	238
203375_s.at	5.442	5.684	-2.82	239
202787_s.at	6.923	6.791	2.82	240
214196_s.at	6.078	5.895	2.82	241
204099.at	6.901	6.760	2.82	242
238044.at	6.305	6.181	2.82	243
210484_s.at	8.279	8.080	2.81	244
205627.at	9.210	8.987	2.81	245
209453.at	6.310	6.213	2.81	246
210638_s.at	9.570	9.744	-2.81	247
202142.at	6.063	6.200	-2.80	248
227667.at	6.164	6.054	2.80	249
212516.at	7.235	7.105	2.80	250
226906_s.at	9.033	8.845	2.80	251
201954.at	10.394	10.290	2.80	252
229215.at	6.070	5.942	2.80	253
202900_s.at	6.584	6.782	-2.80	254
224611_s.at	8.087	7.971	2.80	255
222468.at	7.852	7.714	2.80	256
201749.at	6.652	6.552	2.80	257
204212.at	6.766	6.676	2.79	258

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
235383_at	6.168	6.023	2.79	259
1569257_at	6.964	6.755	2.79	260
212846_at	7.225	7.453	-2.79	261
228230_at	6.710	6.559	2.79	262
216676_x_at	5.800	5.626	2.79	263
203729_at	10.782	10.686	2.79	264
220494_s_at	5.786	5.388	2.79	265
227069_at	6.504	6.277	2.79	266
207907_at	8.181	8.058	2.78	267
217078_s_at	8.815	8.666	2.78	268
226715_at	5.688	5.829	-2.78	269
240166_x_at	6.688	6.590	2.78	270
236869_at	7.514	7.246	2.78	271
205419_at	7.202	7.421	-2.78	272
217844_at	9.248	9.135	2.78	273
227897_at	7.455	7.155	2.78	274
201392_s_at	8.356	8.119	2.78	275
228113_at	8.227	8.111	2.78	276
217521_at	7.038	6.888	2.78	277
217662_x_at	5.840	5.672	2.78	278
219256_s_at	5.797	5.647	2.78	279
209608_s_at	6.242	6.338	-2.78	280
220954_s_at	6.804	6.630	2.77	281
214866_at	6.295	6.165	2.77	282
223454_at	7.563	7.421	2.77	283
238642_at	8.379	8.108	2.77	284
214472_at	5.951	5.697	2.77	285
240451_at	5.964	5.784	2.77	286
218194_at	7.264	7.417	-2.77	287
212969_x_at	7.294	7.178	2.76	288
201350_at	9.807	9.643	2.76	289
237104_at	8.814	8.614	2.76	290
229364_at	6.076	5.880	2.76	291
243099_at	8.641	8.518	2.76	292
210554_s_at	9.599	9.495	2.76	293
212987_at	8.762	8.948	-2.76	294
215631_s_at	7.460	7.373	2.75	295
215050_x_at	5.891	5.801	2.75	296
221234_s_at	5.931	6.191	-2.75	297
218902_at	7.921	7.787	2.75	298
217911_s_at	5.813	5.999	-2.75	299
233072_at	6.245	6.104	2.75	300
212895_s_at	7.905	7.825	2.75	301
209933_s_at	10.448	10.344	2.75	302
1554316_at	5.884	5.763	2.75	303
216565_x_at	9.670	9.571	2.75	304
200807_s_at	8.500	8.678	-2.74	305
211816_x_at	5.666	5.490	2.74	306
215175_at	5.634	5.388	2.74	307
206335_at	7.485	7.381	2.74	308
203047_at	8.829	8.717	2.74	309
206666_at	8.066	8.325	-2.74	310

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
240656_at	6.360	6.155	2.74	311
1553695_a_at	6.103	6.004	2.74	312
200078_s_at	10.387	10.270	2.74	313
207131_x_at	6.115	5.991	2.74	314
228775_at	6.814	6.958	-2.74	315
202848_s_at	7.783	7.674	2.73	316
213185_at	6.292	6.173	2.73	317
217650_x_at	7.249	7.162	2.73	318
224025_s_at	6.778	6.666	2.73	319
209607_x_at	7.920	7.769	2.73	320
217473_x_at	6.600	6.482	2.73	321
230292_at	9.028	8.882	2.73	322
209534_x_at	8.826	8.690	2.73	323
220036_s_at	6.323	6.225	2.73	324
201401_s_at	7.571	7.426	2.73	325
226445_s_at	5.927	5.804	2.73	326
217763_s_at	10.166	9.969	2.72	327
219183_s_at	9.423	9.294	2.72	328
211943_x_at	12.175	12.328	-2.72	329
244753_at	7.887	7.670	2.72	330
213589_s_at	6.038	5.893	2.72	331
216342_x_at	11.188	11.315	-2.72	332
218487_at	6.229	6.357	-2.72	333
55093_at	6.904	6.821	2.71	334
242554_at	5.974	5.872	2.71	335
200852_x_at	9.560	9.417	2.71	336
202795_x_at	8.885	8.806	2.71	337
205684_s_at	5.490	5.688	-2.71	338
218272_at	7.837	7.702	2.71	339
215273_s_at	7.878	7.753	2.71	340
203388_at	9.891	9.768	2.71	341
231406_at	7.223	7.080	2.71	342
224785_at	6.673	6.542	2.71	343
205068_s_at	8.952	8.808	2.70	344
205546_s_at	8.576	8.436	2.70	345
1554516_at	5.734	5.587	2.70	346
226400_at	9.669	9.468	2.70	347
211037_s_at	6.910	6.762	2.70	348
205180_s_at	8.637	8.484	2.70	349
222143_s_at	8.169	8.067	2.70	350
200660_at	11.713	11.577	2.70	351
241722_x_at	9.735	9.578	2.69	352
202178_at	6.864	6.762	2.69	353
232543_x_at	10.404	10.256	2.69	354
225175_s_at	8.556	8.444	2.69	355
202788_at	9.188	9.098	2.69	356
1558732_at	7.544	7.344	2.69	357
212607_at	5.842	5.985	-2.69	358
212413_at	6.146	6.347	-2.69	359
224864_at	8.008	7.912	2.69	360
226168_at	7.146	7.048	2.69	361
223533_at	6.451	6.596	-2.69	362

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
212041_at	10.850	10.701	2.69	363
213517_at	7.763	7.536	2.69	364
227618_at	8.368	8.204	2.69	365
244218_at	7.284	7.098	2.69	366
228109_at	5.605	5.755	-2.68	367
219358_s_at	6.780	6.660	2.68	368
200695_at	7.406	7.291	2.68	369
226842_at	8.523	8.417	2.68	370
241353_s_at	6.845	6.724	2.68	371
40225_at	8.674	8.580	2.68	372
220404_at	8.469	8.287	2.68	373
202191_s_at	7.955	7.779	2.68	374
211615_s_at	6.096	6.297	-2.68	375
233385_x_at	7.880	7.761	2.68	376
219566_at	6.902	6.783	2.67	377
225883_at	9.054	8.892	2.67	378
1556472_s_at	5.808	5.972	-2.67	379
240960_at	7.150	6.934	2.67	380
214327_x_at	11.953	12.089	-2.67	381
224890_s_at	9.145	9.009	2.67	382
208540_x_at	11.344	11.247	2.66	383
227294_at	6.105	6.221	-2.66	384
233510_s_at	8.038	7.878	2.66	385
207783_x_at	10.136	10.252	-2.66	386
211299_s_at	7.388	7.152	2.65	387
225463_x_at	6.612	6.753	-2.65	388
226328_at	6.192	6.104	2.65	389
224676_at	8.466	8.277	2.65	390
215227_x_at	6.718	6.838	-2.65	391
200805_at	7.625	7.541	2.65	392
226074_at	8.137	8.010	2.65	393
223124_s_at	8.971	9.167	-2.65	394
208804_s_at	8.436	8.588	-2.64	395
201846_s_at	9.244	9.032	2.64	396
221800_s_at	6.601	6.495	2.64	397
202122_s_at	8.422	8.276	2.64	398
221293_s_at	7.105	6.996	2.64	399
212974_at	6.411	6.235	2.64	400
207872_s_at	6.161	6.055	2.64	401
231908_at	5.914	5.768	2.64	402
222482_at	6.335	6.503	-2.64	403
202646_s_at	8.030	8.190	-2.64	404
221432_s_at	6.464	6.338	2.64	405
218714_at	7.383	7.252	2.63	406
230543_at	5.255	4.781	2.63	407
214124_x_at	6.140	6.291	-2.63	408
203823_at	7.289	7.163	2.63	409
1559436_x_at	9.750	9.544	2.63	410
233056_x_at	7.666	7.433	2.63	411
212669_at	7.676	7.513	2.63	412
210011_s_at	6.958	6.863	2.62	413
232094_at	6.185	5.875	2.62	414

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
220054_at	5.873	5.966	-2.62	415
240282_at	7.090	6.928	2.62	416
1558515_at	6.144	5.956	2.62	417
201623_s_at	9.012	9.249	-2.62	418
239023_at	7.236	7.017	2.62	419
200994_at	6.234	6.441	-2.62	420
211100_x_at	8.819	8.657	2.62	421
204046_at	8.177	8.068	2.62	422
233315_at	6.145	6.038	2.62	423
201998_at	8.267	8.409	-2.62	424
219290_x_at	7.079	6.917	2.62	425
223015_at	7.753	7.961	-2.62	426
203175_at	11.189	11.048	2.61	427
211101_x_at	8.823	8.657	2.61	428
209002_s_at	7.753	7.660	2.61	429
217977_at	11.601	11.481	2.61	430
1559530_at	6.116	5.963	2.61	431
1556646_at	6.531	6.397	2.61	432
224706_at	8.504	8.399	2.61	433
218302_at	7.828	7.685	2.61	434
201868_s_at	7.145	6.887	2.61	435
212869_x_at	12.440	12.591	-2.61	436
225641_at	6.704	6.623	2.60	437
217848_s_at	7.864	8.038	-2.60	438
236226_at	5.640	5.834	-2.60	439
218555_at	6.631	6.508	2.60	440
211385_x_at	8.110	7.985	2.60	441
208610_s_at	8.196	8.025	2.60	442
224833_at	9.995	10.199	-2.60	443
223265_at	6.387	6.279	2.60	444
206976_s_at	6.366	6.625	-2.60	445
212772_s_at	6.026	5.899	2.60	446
230707_at	10.113	9.936	2.60	447
1557688_at	8.962	8.771	2.59	448
217673_x_at	9.526	9.609	-2.59	449
212770_at	8.192	8.004	2.59	450
213311_s_at	8.991	8.862	2.59	451
218624_s_at	7.149	7.007	2.59	452
209870_s_at	7.438	7.518	-2.59	453
1557228_at	7.435	7.310	2.59	454
227252_at	8.853	8.745	2.59	455
202155_s_at	8.959	8.844	2.59	456
226725_at	7.173	7.352	-2.58	457
206551_x_at	6.311	6.120	2.58	458
223531_x_at	6.628	6.759	-2.58	459
219447_s_at	6.439	6.362	2.58	460
226405_s_at	7.210	7.071	2.58	461
218787_x_at	6.303	6.219	2.58	462
213564_x_at	11.021	11.163	-2.58	463
201909_at	7.634	6.802	2.58	464
1569345_at	6.234	6.065	2.58	465
233167_at	5.844	5.629	2.58	466

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
201479_at	7.403	7.555	-2.58	467
225382_at	5.672	5.831	-2.57	468
232946_s_at	7.609	7.493	2.57	469
1566959_at	7.775	7.515	2.57	470
211097_s_at	6.152	6.037	2.57	471
230489_at	6.861	6.997	-2.57	472
201087_at	8.579	8.469	2.56	473
238423_at	7.576	7.482	2.56	474
1558327_at	7.347	7.210	2.56	475
211971_s_at	6.843	7.089	-2.56	476
202556_s_at	7.056	6.950	2.56	477
240271_at	7.688	7.507	2.56	478
223609_at	6.811	6.658	2.56	479
208583_x_at	7.043	6.957	2.56	480
237591_at	7.903	7.741	2.56	481
240862_at	7.495	7.354	2.56	482
229021_at	7.687	7.571	2.56	483
202455_at	7.819	7.716	2.56	484
214218_s_at	5.899	6.601	-2.56	485
59375_at	6.432	6.278	2.56	486
243329_at	6.417	6.229	2.56	487
202255_s_at	7.599	7.492	2.55	488
203907_s_at	8.422	8.321	2.55	489
239464_at	7.104	6.945	2.55	490
236381_s_at	6.373	6.227	2.55	491
239780_at	6.037	5.902	2.55	492
1563621_at	5.824	5.621	2.55	493
223828_s_at	5.831	5.939	-2.55	494
57739_at	7.552	7.427	2.55	495
224284_x_at	8.381	8.195	2.55	496
241940_at	6.590	6.324	2.55	497
1559128_at	6.304	6.169	2.55	498
205118_at	7.183	6.980	2.55	499
243748_at	7.664	7.440	2.55	500
223754_at	6.462	6.630	-2.54	501
212947_at	6.972	6.863	2.54	502
214153_at	6.999	6.846	2.54	503
212975_at	8.589	8.455	2.54	504
219981_x_at	7.073	6.943	2.54	505
213607_x_at	8.482	8.331	2.54	506
201930_at	7.114	7.258	-2.54	507
212492_s_at	8.937	8.764	2.54	508
213596_at	7.012	6.850	2.54	509
1554501_at	7.500	7.355	2.54	510
222423_at	5.832	6.140	-2.54	511
243147_x_at	7.553	7.355	2.54	512
207163_s_at	7.778	7.637	2.54	513
235421_at	6.989	6.769	2.54	514
222405_at	6.454	6.586	-2.54	515
210598_at	6.368	6.209	2.54	516
207979_s_at	7.202	7.368	-2.54	517
226854_at	6.016	5.889	2.54	518

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
227677_at	9.537	9.406	2.54	519
238996_x_at	7.557	7.424	2.53	520
219105_x_at	6.071	5.948	2.53	521
218298_s_at	8.724	8.612	2.53	522
230085_at	7.595	7.380	2.53	523
223331_s_at	6.022	6.137	-2.53	524
237568_at	7.941	7.742	2.53	525
203912_s_at	7.829	7.733	2.53	526
214780_s_at	7.662	7.581	2.53	527
206723_s_at	8.608	8.480	2.53	528
225234_at	9.109	8.940	2.53	529
241268_x_at	6.822	6.674	2.53	530
200919_at	10.096	9.971	2.53	531
207671_s_at	6.415	6.300	2.53	532
205119_s_at	12.524	12.400	2.53	533
1556049_at	6.516	6.305	2.52	534
228443_s_at	7.461	7.363	2.52	535
239179_at	7.800	7.640	2.52	536
225725_at	5.433	5.623	-2.52	537
223462_at	6.180	6.087	2.52	538
206656_s_at	9.753	9.618	2.52	539
200710_at	8.101	7.990	2.52	540
237176_at	8.536	8.319	2.52	541
209083_at	11.386	11.241	2.52	542
233334_x_at	7.289	7.209	2.52	543
241632_x_at	6.357	6.186	2.51	544
244023_at	8.883	8.709	2.51	545
220352_x_at	6.686	6.530	2.51	546
221006_s_at	8.340	8.168	2.51	547
228499_at	7.632	7.511	2.51	548
235984_at	7.288	7.111	2.51	549
203136_at	9.016	8.893	2.51	550
52169_at	8.437	8.361	2.51	551
212378_at	6.014	6.165	-2.51	552
218612_s_at	6.468	6.377	2.51	553
211285_s_at	7.126	7.289	-2.51	554
212910_at	6.168	6.292	-2.50	555
1557270_at	6.219	6.064	2.50	556
241824_at	6.632	6.349	2.50	557
239695_at	6.787	6.607	2.50	558
241742_at	7.622	7.476	2.50	559
201162_at	6.904	6.784	2.50	560
211605_s_at	6.509	6.373	2.50	561
201326_at	5.994	6.196	-2.50	562
218918_at	5.642	5.823	-2.50	563
63825_at	8.048	7.850	2.50	564
229548_at	6.643	6.513	2.50	565
200846_s_at	9.771	9.676	2.50	566
237218_at	6.237	6.097	2.50	567
117_at	8.292	8.130	2.50	568
227129_x_at	10.108	9.921	2.50	569
221726_at	9.135	9.277	-2.50	570

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
60471_at	8.789	8.622	2.50	571
201047_x_at	7.726	7.921	-2.49	572
201234_at	9.280	9.187	2.49	573
203615_x_at	9.216	9.071	2.49	574
230805_at	7.434	7.291	2.49	575
212219_at	6.869	7.009	-2.49	576
209282_at	7.947	7.819	2.49	577
203733_at	6.244	6.330	-2.49	578
219298_at	6.001	5.874	2.49	579
225582_at	8.958	8.853	2.49	580
238619_at	9.631	9.479	2.49	581
205353_s_at	6.882	7.006	-2.49	582
221512_at	7.311	7.150	2.49	583
214366_s_at	7.834	7.642	2.49	584
236571_at	7.138	6.948	2.49	585
229593_at	8.369	8.193	2.49	586
242377_x_at	6.279	6.166	2.48	587
233816_at	6.586	6.368	2.48	588
232652_x_at	7.131	7.015	2.48	589
200717_x_at	11.047	11.382	-2.48	590
202791_s_at	6.389	6.317	2.48	591
219807_x_at	7.751	7.640	2.48	592
205645_at	7.511	7.369	2.48	593
210569_s_at	6.392	6.282	2.48	594
235954_at	6.959	6.765	2.48	595
1560706_at	8.193	8.003	2.48	596
200752_s_at	6.801	6.679	2.48	597
208947_s_at	6.217	6.107	2.48	598
243537_at	6.219	6.071	2.48	599
212242_at	9.871	9.769	2.48	600
208919_s_at	10.021	9.881	2.48	601
228638_at	6.093	5.877	2.48	602
219165_at	7.877	7.772	2.48	603
219667_s_at	6.293	6.530	-2.48	604
204265_s_at	11.021	10.911	2.48	605
1555303_at	8.003	7.817	2.48	606
206169_x_at	6.559	6.323	2.48	607
201381_x_at	7.636	7.761	-2.48	608
205639_at	8.893	8.773	2.47	609
230383_x_at	6.633	6.847	-2.47	610
227696_at	5.924	6.050	-2.47	611
202970_at	5.811	5.936	-2.47	612
201919_at	6.210	6.435	-2.47	613
219690_at	7.923	7.827	2.47	614
220147_s_at	6.847	7.027	-2.47	615
56197_at	9.363	9.447	-2.47	616
205254_x_at	6.308	6.465	-2.47	617
1554503_a_at	6.635	6.465	2.47	618
203523_at	10.074	9.925	2.47	619
1555154_a_at	5.863	5.712	2.47	620
210483_at	7.556	7.388	2.47	621
200050_at	6.678	6.915	-2.47	622

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
226442_at	7.789	7.617	2.46	623
212956_at	6.233	6.510	-2.46	624
209110_s_at	7.862	7.750	2.46	625
244357_at	8.434	8.174	2.46	626
233867_at	8.536	8.343	2.46	627
212407_at	6.608	6.766	-2.46	628
234196_at	6.823	6.590	2.46	629
217200_x_at	6.333	6.267	2.46	630
218978_s_at	8.074	7.892	2.46	631
203508_at	9.702	9.610	2.46	632
210589_s_at	7.025	6.924	2.46	633
214992_s_at	6.115	6.004	2.46	634
223852_s_at	7.922	7.815	2.46	635
213327_s_at	5.439	5.599	-2.46	636
216835_s_at	6.738	6.640	2.46	637
214486_x_at	7.820	7.704	2.46	638
227726_at	7.694	7.545	2.45	639
231644_at	7.026	6.812	2.45	640
208687_x_at	10.089	10.214	-2.45	641
202741_at	7.753	8.028	-2.45	642
230388_s_at	6.481	6.772	-2.45	643
228630_at	5.459	5.628	-2.45	644
203727_at	6.852	6.775	2.45	645
222429_at	7.737	7.611	2.45	646
1569150_x_at	5.814	5.655	2.45	647
213300_at	8.168	8.019	2.45	648
219788_at	7.811	7.672	2.45	649
203049_s_at	5.535	5.779	-2.45	650
1552634_a_at	5.604	5.753	-2.45	651
226840_at	8.477	8.355	2.45	652
203921_at	7.263	7.155	2.45	653
1570078_a_at	7.121	6.981	2.45	654
1560199_x_at	6.860	6.679	2.45	655
222218_s_at	9.973	9.832	2.45	656
225174_at	5.547	5.807	-2.44	657
242398_x_at	5.755	5.627	2.44	658
215159_s_at	6.655	6.513	2.44	659
202746_at	7.359	7.619	-2.44	660
204903_x_at	7.824	7.741	2.44	661
201166_s_at	7.692	7.889	-2.44	662
225387_at	7.603	7.798	-2.44	663
229595_at	5.990	6.081	-2.44	664
38269_at	9.053	8.930	2.44	665
224451_x_at	11.185	11.054	2.44	666
217715_x_at	6.912	6.748	2.44	667
223008_s_at	6.941	7.132	-2.44	668
244349_at	6.729	6.504	2.43	669
208200_at	5.979	5.730	2.43	670
241968_at	5.491	5.636	-2.43	671
200045_at	7.427	7.538	-2.43	672
243256_at	8.555	8.396	2.43	673
241610_x_at	6.493	6.068	2.43	674

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
202420_s.at	6.888	6.983	-2.43	675
233964.at	5.246	4.967	2.43	676
240695.at	7.071	6.863	2.43	677
203104.at	7.886	7.751	2.43	678
204164.at	8.216	8.091	2.43	679
233314.at	6.750	6.564	2.43	680
227520.at	5.223	5.445	-2.43	681
209354.at	7.578	7.510	2.43	682
202125_s.at	6.764	6.953	-2.43	683
217363_x.at	6.853	6.932	-2.42	684
209221_s.at	7.560	7.442	2.42	685
209065.at	5.947	6.074	-2.42	686
224909_s.at	9.888	9.754	2.42	687
225050.at	7.579	7.710	-2.42	688
209411_s.at	7.590	7.486	2.42	689
215372_x.at	6.617	6.514	2.42	690
233775_x.at	6.463	6.276	2.42	691
200871_s.at	12.197	12.111	2.42	692
203530_s.at	7.724	7.637	2.42	693
232253.at	8.549	8.460	2.42	694
243514.at	6.704	6.544	2.42	695
203814_s.at	8.619	8.414	2.42	696
244556.at	7.460	7.301	2.42	697
209001_s.at	8.761	8.929	-2.41	698
200708.at	7.258	7.393	-2.41	699
208450.at	6.742	6.966	-2.41	700
206600_s.at	6.946	6.818	2.41	701
237077.at	6.409	6.222	2.41	702
223637_s.at	6.891	6.793	2.41	703
214181_x.at	9.688	9.554	2.41	704
206437.at	8.893	8.789	2.41	705
203633.at	6.835	6.697	2.41	706
220320.at	7.105	7.012	2.41	707
229770.at	9.606	9.489	2.41	708
224187_x.at	11.054	11.165	-2.41	709
214039_s.at	5.568	5.767	-2.41	710
226707.at	7.413	7.301	2.41	711
226528.at	5.475	5.681	-2.40	712
207677_s.at	10.430	10.270	2.40	713
201833.at	6.021	6.217	-2.40	714
202809_s.at	6.897	6.780	2.40	715
201155_s.at	7.457	7.355	2.40	716
204425.at	8.639	8.528	2.40	717
227134.at	9.091	8.979	2.40	718
239213.at	7.213	7.093	2.40	719
239901.at	8.252	8.130	2.40	720
240154.at	9.365	9.216	2.40	721
221267_s.at	9.562	9.432	2.40	722
224849.at	7.257	7.355	-2.40	723
213617_s.at	6.391	6.529	-2.40	724
34206.at	7.842	7.676	2.40	725
243931.at	8.384	8.238	2.40	726

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
219083_at	5.668	5.845	-2.40	727
234644_x_at	9.425	9.261	2.40	728
205672_at	5.680	5.813	-2.40	729
31837_at	8.108	8.038	2.40	730
242761_s_at	5.486	5.666	-2.40	731
203271_s_at	7.488	7.398	2.40	732
232478_at	5.616	5.462	2.40	733
213812_s_at	9.214	9.046	2.40	734
229164_s_at	9.774	9.662	2.40	735
208833_s_at	8.460	8.607	-2.39	736
224573_at	11.538	11.460	2.39	737
220476_s_at	6.641	6.504	2.39	738
230000_at	8.102	7.912	2.39	739
1559201_a_at	7.038	6.889	2.39	740
221541_at	9.144	8.974	2.39	741
53987_at	6.190	6.317	-2.39	742
218667_at	7.435	7.534	-2.39	743
226401_at	6.272	6.183	2.39	744
227131_at	7.923	7.806	2.39	745
218746_at	7.097	6.996	2.39	746
206881_s_at	7.667	7.458	2.39	747
205781_at	6.436	6.321	2.39	748
221755_at	6.871	6.705	2.39	749
210859_x_at	7.424	7.327	2.39	750
210785_s_at	9.594	9.468	2.39	751
206056_x_at	7.068	6.921	2.39	752
236338_at	6.506	6.293	2.39	753
237318_at	5.808	5.637	2.38	754
244842_x_at	6.461	6.346	2.38	755
211316_x_at	9.351	9.204	2.38	756
229563_s_at	12.812	12.902	-2.38	757
201369_s_at	7.616	7.408	2.38	758
200766_at	6.509	6.391	2.38	759
239408_at	7.572	7.410	2.38	760
243860_at	7.127	7.024	2.38	761
231108_at	5.827	5.631	2.37	762
212333_at	5.627	5.772	-2.37	763
224261_at	8.289	8.169	2.37	764
223092_at	7.310	7.445	-2.37	765
224303_x_at	6.141	6.043	2.37	766
232952_at	5.430	5.110	2.37	767
214746_s_at	8.269	8.105	2.37	768
200631_s_at	8.934	9.106	-2.37	769
224880_at	7.503	7.691	-2.37	770
224898_at	5.686	5.888	-2.37	771
49878_at	8.289	8.240	2.37	772
202180_s_at	9.022	8.910	2.37	773
203692_s_at	7.303	7.190	2.37	774
220964_s_at	9.009	8.911	2.36	775
215063_x_at	6.979	6.854	2.36	776
205707_at	8.842	8.737	2.36	777
227261_at	6.373	6.622	-2.36	778

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
202073_at	5.912	6.121	-2.36	779
205006_s_at	5.433	5.606	-2.36	780
232030_at	5.614	5.377	2.36	781
213733_at	10.776	10.662	2.36	782
1554952_s_at	7.660	7.528	2.36	783
207890_s_at	9.752	9.582	2.36	784
218422_s_at	7.657	7.881	-2.36	785
204786_s_at	8.199	8.032	2.36	786
228298_at	8.030	8.183	-2.36	787
211102_s_at	6.380	6.196	2.36	788
204174_at	11.785	11.655	2.36	789
208699_x_at	9.970	9.852	2.36	790
212505_s_at	8.248	8.155	2.36	791
224321_at	7.624	7.241	2.36	792
219862_s_at	7.966	7.823	2.36	793
232215_x_at	9.234	9.055	2.36	794
225618_at	6.793	6.680	2.35	795
211883_x_at	6.143	6.025	2.35	796
218403_at	5.578	5.754	-2.35	797
214807_at	6.682	6.501	2.35	798
241630_at	5.715	5.576	2.35	799
221155_x_at	6.839	6.688	2.35	800
223182_s_at	6.217	6.048	2.35	801
213418_at	9.726	9.571	2.35	802
222721_at	6.524	6.374	2.35	803
209275_s_at	7.589	7.494	2.35	804
222099_s_at	6.459	6.548	-2.35	805
225755_at	6.845	6.730	2.35	806
204495_s_at	7.642	7.543	2.35	807
202621_at	7.655	7.532	2.35	808
203198_at	6.158	6.036	2.35	809
234135_x_at	6.310	6.214	2.35	810
217992_s_at	9.676	9.496	2.35	811
208922_s_at	8.257	8.144	2.35	812
218098_at	5.543	5.766	-2.34	813
202111_at	6.434	6.343	2.34	814
221882_s_at	8.204	8.081	2.34	815
238610_s_at	5.870	5.747	2.34	816
230987_at	7.849	7.646	2.34	817
200972_at	6.427	6.606	-2.34	818
216745_x_at	6.401	6.255	2.34	819
213261_at	8.350	8.220	2.34	820
239474_at	7.684	7.476	2.34	821
1558299_at	7.748	7.565	2.34	822
221790_s_at	8.369	8.474	-2.34	823
205469_s_at	6.587	6.487	2.34	824
219394_at	8.201	8.061	2.34	825
208454_s_at	7.606	7.498	2.34	826
232876_at	7.881	7.722	2.34	827
212359_s_at	6.899	6.778	2.34	828
227501_at	6.888	6.733	2.34	829
200953_s_at	7.890	8.071	-2.34	830

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
222024_s_at	9.194	9.050	2.34	831
214501_s_at	9.412	9.282	2.34	832
217297_s_at	7.003	6.849	2.33	833
204166_at	6.820	6.636	2.33	834
224648_at	6.026	6.267	-2.33	835
225321_s_at	8.246	8.114	2.33	836
219460_s_at	8.448	8.359	2.33	837
231222_at	7.061	6.953	2.33	838
216894_x_at	5.761	5.607	2.33	839
224936_at	8.689	8.816	-2.33	840
217833_at	5.785	6.031	-2.33	841
210244_at	7.335	7.171	2.33	842
218023_s_at	9.136	9.022	2.33	843
236191_at	5.996	5.896	2.33	844
211121_s_at	6.278	6.203	2.33	845
212414_s_at	8.650	8.803	-2.33	846
221745_at	6.864	6.707	2.33	847
217529_at	6.473	6.316	2.33	848
243196_s_at	6.298	6.197	2.33	849
200915_x_at	6.923	7.197	-2.33	850
228487_s_at	6.256	6.045	2.33	851
225869_s_at	6.356	6.196	2.33	852
207674_at	6.213	5.992	2.33	853
203089_s_at	6.657	6.543	2.33	854
240168_at	5.829	5.650	2.33	855
217915_s_at	5.139	5.387	-2.33	856
208490_x_at	5.764	5.652	2.33	857
226996_at	6.121	6.275	-2.33	858
240665_at	9.482	9.315	2.33	859
208980_s_at	12.274	12.188	2.33	860
225613_at	5.482	5.643	-2.32	861
36554_at	6.910	6.798	2.32	862
219173_at	5.833	5.684	2.32	863
211168_s_at	7.165	7.068	2.32	864
219999_at	6.631	6.521	2.32	865
205324_s_at	6.405	6.558	-2.32	866
200027_at	8.279	8.468	-2.32	867
230872_s_at	5.899	5.803	2.32	868
228599_at	5.498	5.724	-2.32	869
222670_s_at	8.055	7.941	2.32	870
204949_at	11.338	11.214	2.32	871
226171_at	5.837	5.726	2.32	872
221558_s_at	8.718	8.902	-2.32	873
221875_x_at	12.120	12.040	2.32	874
243442_x_at	7.196	7.049	2.32	875
237640_at	5.798	5.641	2.32	876
224754_at	10.040	9.897	2.32	877
217456_x_at	9.730	9.660	2.32	878
219593_at	6.876	6.746	2.32	879
217654_at	5.993	5.842	2.32	880
203066_at	10.501	10.378	2.32	881
242646_at	8.028	7.910	2.32	882

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
201306_s_at	10.539	10.650	-2.32	883
227698_s_at	6.756	6.648	2.32	884
50221_at	8.459	8.340	2.32	885
221246_x_at	6.410	6.579	-2.31	886
203332_s_at	8.651	8.557	2.31	887
239188_at	6.621	6.471	2.31	888
210564_x_at	7.988	7.842	2.31	889
209093_s_at	6.408	6.290	2.31	890
225499_at	6.819	6.699	2.31	891
205998_x_at	6.519	6.409	2.31	892
202317_s_at	8.147	8.055	2.31	893
213669_at	5.889	5.796	2.31	894
212059_s_at	6.195	6.095	2.31	895
213159_at	6.669	6.531	2.31	896
234072_at	5.837	5.670	2.31	897
213431_x_at	6.616	6.518	2.31	898
210279_at	7.253	7.429	-2.31	899
226968_at	6.971	6.849	2.31	900
203973_s_at	10.558	10.440	2.31	901
226207_at	6.002	5.893	2.31	902
215404_x_at	6.879	6.703	2.30	903
206574_s_at	6.759	6.640	2.30	904
232726_at	6.938	6.743	2.30	905
232145_at	7.416	7.264	2.30	906
207540_s_at	9.045	8.928	2.30	907
39891_at	6.897	6.832	2.30	908
1553978_at	7.654	7.543	2.30	909
244197_x_at	8.630	8.449	2.30	910
236291_at	6.140	5.987	2.30	911
206700_s_at	4.515	4.061	2.30	912
205033_s_at	11.198	10.836	2.30	913
205382_s_at	9.340	9.075	2.30	914
204806_x_at	10.070	9.972	2.30	915
242312_x_at	5.865	5.591	2.30	916
239600_at	8.044	7.870	2.30	917
221891_x_at	11.047	11.167	-2.30	918
227906_s_at	5.973	5.883	2.30	919
203387_s_at	5.614	5.854	-2.30	920
218143_s_at	8.339	8.261	2.30	921
224893_at	7.184	7.372	-2.30	922
231597_x_at	6.888	6.221	2.30	923
201810_s_at	8.137	8.040	2.30	924
202913_at	6.412	6.323	2.30	925
208579_x_at	7.999	7.868	2.30	926
232266_x_at	9.289	9.054	2.30	927
239223_s_at	6.312	6.197	2.30	928
223141_at	7.023	6.963	2.30	929
209818_s_at	5.869	5.937	-2.30	930
242413_at	6.085	5.964	2.30	931
208787_at	6.691	6.961	-2.30	932
212024_x_at	8.108	8.031	2.30	933
227465_at	7.732	7.630	2.30	934

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
206809_s.at	6.529	6.663	-2.30	935
239633.at	5.912	5.767	2.30	936
200965_s.at	8.698	8.842	-2.30	937
233358.at	6.464	6.329	2.29	938
203045.at	9.508	9.370	2.29	939
206853_s.at	5.511	5.683	-2.29	940
220711.at	6.969	6.854	2.29	941
203974.at	7.131	7.273	-2.29	942
225454.at	5.904	5.783	2.29	943
226254_s.at	6.571	6.761	-2.29	944
216984_x.at	7.196	7.338	-2.29	945
215633_x.at	9.648	9.534	2.29	946
225260_s.at	6.074	6.242	-2.29	947
217908_s.at	6.962	7.112	-2.29	948
212720.at	5.725	5.951	-2.29	949
228452.at	5.950	6.063	-2.29	950
226849.at	7.207	7.109	2.29	951
227858.at	5.945	5.812	2.29	952
223346.at	5.843	5.750	2.29	953
211582_x.at	9.699	9.578	2.29	954
230142_s.at	6.948	6.771	2.29	955
205370_x.at	9.935	9.772	2.29	956
215373_x.at	7.852	7.637	2.28	957
1554553_s.at	7.020	6.920	2.28	958
239364.at	5.839	5.663	2.28	959
221864.at	7.829	7.736	2.28	960
226266.at	6.977	6.801	2.28	961
232867.at	6.192	6.086	2.28	962
213060_s.at	5.615	5.761	-2.28	963
214911_s.at	9.189	9.050	2.28	964
209876.at	5.870	5.717	2.28	965
219378.at	6.045	6.236	-2.28	966
205315_s.at	7.046	6.900	2.28	967
226588.at	5.538	5.663	-2.28	968
225373.at	10.753	10.630	2.28	969
211452_x.at	10.688	10.586	2.28	970
209831_x.at	7.761	7.666	2.28	971
213233_s.at	6.932	7.092	-2.28	972
242990.at	6.435	6.237	2.28	973
216229_x.at	7.427	7.185	2.28	974
218840_s.at	6.986	6.884	2.28	975
236107.at	6.431	6.301	2.27	976
222217_s.at	6.375	6.235	2.27	977
243934.at	7.092	6.963	2.27	978
207205.at	7.558	7.462	2.27	979
212692_s.at	6.523	6.647	-2.27	980
224806.at	9.627	9.496	2.27	981
225302.at	5.856	6.092	-2.27	982
231861.at	7.326	7.183	2.27	983
208485_x.at	8.630	8.477	2.27	984
205963_s.at	6.773	6.848	-2.27	985
235085.at	7.853	7.981	-2.27	986

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
237563_s_at	6.602	6.488	2.27	987
221970_s_at	6.576	6.725	-2.27	988
236475_at	7.051	6.869	2.27	989
202117_at	9.131	9.051	2.27	990
234159_at	6.083	5.955	2.26	991
217884_at	6.241	6.343	-2.26	992
228238_at	5.756	5.854	-2.26	993
232141_at	7.310	7.093	2.26	994
212465_at	6.920	6.996	-2.26	995
229957_at	6.711	6.618	2.26	996
213480_at	6.153	6.305	-2.26	997
201687_s_at	6.615	6.824	-2.26	998
220739_s_at	7.413	7.275	2.26	999
219452_at	9.430	9.310	2.26	1000
225858_s_at	7.646	7.526	2.26	1001
233371_at	5.939	6.192	-2.26	1002
223121_s_at	5.484	5.627	-2.26	1003
213527_s_at	6.671	6.582	2.26	1004
221506_s_at	6.855	6.756	2.25	1005
237864_at	6.304	6.183	2.25	1006
228089_x_at	8.269	8.184	2.25	1007
213093_at	5.812	6.019	-2.25	1008
240638_at	6.615	6.464	2.25	1009
225680_at	6.610	6.526	2.25	1010
227060_at	7.391	7.302	2.25	1011
210789_x_at	7.893	7.782	2.25	1012
223195_s_at	7.050	6.952	2.25	1013
226758_at	6.203	6.333	-2.25	1014
200034_s_at	10.757	10.909	-2.25	1015
227871_at	5.462	5.609	-2.25	1016
212717_at	8.105	8.026	2.25	1017
1557553_at	6.420	6.255	2.25	1018
216863_s_at	6.202	6.315	-2.25	1019
218215_s_at	6.610	6.465	2.25	1020
1555643_s_at	7.374	7.136	2.25	1021
218910_at	5.915	5.828	2.25	1022
209394_at	7.633	7.559	2.24	1023
235993_at	5.627	5.779	-2.24	1024
212313_at	7.644	7.739	-2.24	1025
211521_s_at	6.608	6.484	2.24	1026
208667_s_at	6.771	6.995	-2.24	1027
233929_x_at	7.432	7.317	2.24	1028
214290_s_at	10.638	10.517	2.24	1029
213062_at	7.070	7.184	-2.24	1030
221011_s_at	8.385	8.542	-2.24	1031
213097_s_at	5.665	5.801	-2.24	1032
235568_at	7.929	7.764	2.24	1033
227891_s_at	7.758	7.602	2.24	1034
1557051_s_at	7.041	6.839	2.24	1035
217809_at	7.517	7.631	-2.24	1036
201119_s_at	9.727	9.638	2.24	1037
200933_x_at	13.022	13.110	-2.24	1038

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
221798_x_at	13.112	13.166	-2.24	1039
229293_at	7.172	7.081	2.24	1040
211862_x_at	8.603	8.452	2.23	1041
203087_s_at	5.659	5.878	-2.23	1042
243469_at	7.896	7.679	2.23	1043
218155_x_at	7.273	7.104	2.23	1044
200998_s_at	8.272	8.109	2.23	1045
239701_at	6.398	6.197	2.23	1046
211933_s_at	8.504	8.609	-2.23	1047
222145_at	5.906	5.703	2.23	1048
210036_s_at	6.472	6.586	-2.23	1049
1559502_s_at	8.536	8.391	2.23	1050
205462_s_at	6.812	6.680	2.23	1051
212042_x_at	10.496	10.793	-2.23	1052
207571_x_at	9.580	9.470	2.23	1053
209623_at	6.807	6.923	-2.23	1054
58780_s_at	8.001	7.827	2.23	1055
212642_s_at	7.563	7.635	-2.23	1056
224630_at	6.648	6.757	-2.23	1057
212592_at	6.947	7.227	-2.23	1058
213716_s_at	9.841	9.729	2.23	1059
232516_x_at	6.570	6.426	2.23	1060
206687_s_at	10.639	10.526	2.23	1061
214188_at	6.152	6.011	2.23	1062
235478_at	6.034	6.161	-2.23	1063
212360_at	8.345	8.225	2.23	1064
226383_at	5.811	5.963	-2.23	1065
204542_at	6.993	6.854	2.22	1066
202754_at	6.763	6.885	-2.22	1067
212203_x_at	12.430	12.336	2.22	1068
233427_x_at	7.645	7.500	2.22	1069
221718_s_at	9.062	8.939	2.22	1070
208917_x_at	7.622	7.494	2.22	1071
217728_at	11.538	11.412	2.22	1072
212578_x_at	10.259	10.481	-2.22	1073
224333_s_at	7.049	7.115	-2.22	1074
210835_s_at	9.391	9.300	2.22	1075
229803_s_at	7.432	7.282	2.22	1076
223035_s_at	7.427	7.504	-2.22	1077
223163_s_at	6.324	6.406	-2.22	1078
225352_at	6.083	6.279	-2.22	1079
1569396_at	6.090	6.014	2.22	1080
219766_at	6.223	6.144	2.22	1081
226039_at	6.981	7.234	-2.22	1082
219940_s_at	6.844	6.960	-2.22	1083
205322_s_at	7.603	7.514	2.22	1084
224612_s_at	7.845	7.762	2.22	1085
205875_s_at	7.551	7.456	2.22	1086
211806_s_at	8.323	8.170	2.22	1087
210119_at	9.456	9.299	2.22	1088
1557543_at	7.762	7.537	2.22	1089
211939_x_at	11.065	11.154	-2.22	1090

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
224609_at	8.462	8.342	2.22	1091
218791_s_at	5.203	5.414	-2.22	1092
218454_at	10.589	10.463	2.22	1093
235699_at	6.711	6.568	2.22	1094
226178_at	5.599	5.804	-2.21	1095
236625_at	7.064	6.953	2.21	1096
232807_at	5.773	5.685	2.21	1097
236649_at	6.281	6.397	-2.21	1098
224980_at	6.033	5.916	2.21	1099
211317_s_at	8.665	8.502	2.21	1100
238949_at	6.210	5.964	2.21	1101
221476_s_at	10.236	10.392	-2.21	1102
243554_at	6.567	6.446	2.21	1103
234753_x_at	5.531	5.313	2.21	1104
35436_at	6.939	6.794	2.21	1105
1553588_at	13.589	13.641	-2.21	1106
240652_at	7.087	6.916	2.21	1107
48030_i_at	8.964	9.058	-2.21	1108
204214_s_at	8.310	8.219	2.21	1109
224783_at	8.826	8.746	2.21	1110
208752_x_at	9.662	9.798	-2.21	1111
204599_s_at	7.354	7.289	2.21	1112
1570210_x_at	5.940	5.785	2.21	1113
211563_s_at	5.484	5.638	-2.21	1114
207428_x_at	6.755	6.676	2.21	1115
226452_at	5.641	5.836	-2.21	1116
211050_x_at	6.039	5.890	2.21	1117
217552_x_at	6.599	6.474	2.21	1118
239124_at	6.176	6.027	2.21	1119
230027_s_at	6.019	5.894	2.21	1120
219722_s_at	6.183	6.085	2.21	1121
208801_at	7.608	7.810	-2.21	1122
217831_s_at	5.767	5.648	2.21	1123
239027_at	7.442	7.264	2.20	1124
222428_s_at	6.821	6.961	-2.20	1125
232347_x_at	7.036	6.906	2.20	1126
221999_at	6.601	6.483	2.20	1127
215595_x_at	5.854	5.734	2.20	1128
229249_at	5.759	5.573	2.20	1129
201947_s_at	7.266	7.423	-2.20	1130
219939_s_at	8.980	9.104	-2.20	1131
242144_at	5.880	5.723	2.20	1132
218913_s_at	8.092	7.958	2.20	1133
208120_x_at	9.074	8.933	2.20	1134
200744_s_at	9.090	8.998	2.20	1135
241508_at	7.007	6.851	2.20	1136
202666_s_at	5.472	5.644	-2.20	1137
1555952_at	6.147	6.048	2.20	1138
212234_at	6.140	6.232	-2.20	1139
216241_s_at	8.213	8.388	-2.20	1140
240057_at	6.241	6.134	2.20	1141
218695_at	6.220	6.138	2.20	1142

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
201665_x_at	10.544	10.774	-2.20	1143
216858_x_at	7.387	7.132	2.19	1144
220336_s_at	5.708	5.602	2.19	1145
238534_at	8.592	8.457	2.19	1146
224935_at	8.506	8.644	-2.19	1147
210679_x_at	7.509	7.320	2.19	1148
201260_s_at	6.437	6.602	-2.19	1149
224493_x_at	7.368	7.220	2.19	1150
1556988_s_at	5.908	6.013	-2.19	1151
203709_at	6.546	6.456	2.19	1152
208238_x_at	8.230	8.091	2.19	1153
201024_x_at	8.130	8.267	-2.19	1154
224951_at	7.790	7.717	2.19	1155
223096_at	7.798	7.976	-2.19	1156
239021_at	7.175	6.990	2.19	1157
1557966_x_at	5.982	6.109	-2.19	1158
214017_s_at	7.100	6.994	2.19	1159
228532_at	9.627	9.526	2.19	1160
209398_at	7.352	7.213	2.19	1161
204487_s_at	7.535	7.466	2.19	1162
221666_s_at	9.883	9.758	2.19	1163
203012_x_at	12.236	12.295	-2.19	1164
236715_x_at	8.201	8.046	2.19	1165
218607_s_at	6.606	6.734	-2.19	1166
214661_s_at	6.284	6.362	-2.19	1167
213598_at	5.967	6.150	-2.19	1168
204957_at	5.965	6.148	-2.19	1169
234562_x_at	5.370	5.008	2.18	1170
240155_x_at	7.084	6.941	2.18	1171
203110_at	7.797	7.693	2.18	1172
205775_at	5.897	5.973	-2.18	1173
223662_x_at	7.601	7.465	2.18	1174
222158_s_at	6.812	6.694	2.18	1175
223439_at	5.591	5.724	-2.18	1176
206150_at	9.286	9.395	-2.18	1177
215178_x_at	6.889	6.816	2.18	1178
243966_at	7.315	7.219	2.18	1179
204981_at	6.846	6.718	2.18	1180
202602_s_at	5.717	5.906	-2.18	1181
206599_at	5.733	5.636	2.18	1182
228410_at	7.552	7.469	2.18	1183
232077_s_at	7.701	7.550	2.18	1184
203299_s_at	5.908	6.017	-2.18	1185
208052_x_at	7.483	7.382	2.18	1186
227999_at	6.499	6.419	2.18	1187
205010_at	6.106	6.026	2.18	1188
223180_s_at	5.636	5.752	-2.18	1189
202065_s_at	6.187	6.065	2.18	1190
201126_s_at	8.343	8.249	2.18	1191
200065_s_at	10.709	10.637	2.18	1192
208918_s_at	10.360	10.255	2.18	1193
236592_at	7.115	6.922	2.17	1194

continued

Probe	mean PD	mean CTL	Welch t	$ t $ rank
47773_at	6.339	6.251	2.17	1195
225598_at	7.720	7.586	2.17	1196
211454_x_at	8.977	8.788	2.17	1197
212680_x_at	7.377	7.270	2.17	1198
220948_s_at	7.931	8.025	-2.17	1199
241692_at	6.205	6.082	2.17	1200

E GEO cohort composition

E.1 Parkinson arm

Train $n = 293$, test $n = 75$, positive label: IPD.

Cohort	positive	total
TEST	'CONTROL': 40, 'IPD': 30	
TRAINING	'CONTROL': 153, 'IPD': 140	
VALIDATION	'CONTROL': 40, 'IPD': 35	

E.2 Sepsis arm

Train $n = 263$, test $n = 216$, positive label: 1.

Cohort	positive	total
discovery	'0': 194, '1': 69	
validation	'0': 171, '1': 45	

F Pooled GEO sample manifest (complete, 438 samples)

Every sample in the pooled whole-blood matrix with its label and split assignment. GSM accessions resolve at NCBI GEO.

Table 8: Sample manifest.

GSM	Label	Split	Index
GSM2630758	CTL	train	0
GSM2630759	PD	train	1
GSM2630760	PD	train	2
GSM2630761	CTL	train	3
GSM2630762	PD	train	4
GSM2630763	PD	train	5
GSM2630764	PD	val	6
GSM2630765	PD	val	7
GSM2630766	PD	train	8

continued

GSM	Label	Split	Index
GSM2630768	PD	train	9
GSM2630769	CTL	train	10
GSM2630770	PD	train	11
GSM2630771	CTL	train	12
GSM2630772	CTL	train	13
GSM2630773	PD	train	14
GSM2630774	CTL	train	15
GSM2630777	CTL	train	16
GSM2630778	CTL	train	17
GSM2630779	CTL	train	18
GSM2630780	CTL	train	19
GSM2630781	PD	val	20
GSM2630783	CTL	train	21
GSM2630784	CTL	train	22
GSM2630785	CTL	train	23
GSM2630787	CTL	train	24
GSM2630788	CTL	train	25
GSM2630789	CTL	train	26
GSM2630790	CTL	train	27
GSM2630791	PD	val	28
GSM2630793	PD	val	29
GSM2630794	PD	train	30
GSM2630795	PD	train	31
GSM2630799	CTL	train	32
GSM2630800	CTL	train	33
GSM2630801	PD	train	34
GSM2630802	PD	val	35
GSM2630803	CTL	train	36
GSM2630804	PD	train	37
GSM2630805	PD	train	38
GSM2630806	PD	train	39
GSM2630807	PD	train	40
GSM2630810	CTL	train	41
GSM2630812	CTL	train	42
GSM2630813	PD	val	43
GSM2630815	PD	train	44
GSM2630816	CTL	train	45
GSM2630817	PD	train	46
GSM2630818	CTL	val	47
GSM2630821	PD	train	48
GSM2630822	PD	train	49
GSM2630823	PD	train	50
GSM2630824	PD	train	51
GSM2630825	PD	train	52
GSM2630826	PD	train	53
GSM2630827	PD	val	54
GSM2630828	PD	train	55
GSM2630829	PD	train	56
GSM2630830	CTL	train	57
GSM2630835	PD	train	58
GSM2630837	PD	train	59
GSM2630838	PD	val	60

continued

GSM	Label	Split	Index
GSM2630839	PD	val	61
GSM2630840	CTL	train	62
GSM2630841	PD	val	63
GSM2630842	PD	train	64
GSM2630843	CTL	train	65
GSM2630853	CTL	train	66
GSM2630854	PD	train	67
GSM2630855	CTL	train	68
GSM2630856	PD	train	69
GSM2630857	CTL	train	70
GSM2630861	CTL	train	71
GSM2630862	PD	val	72
GSM2630863	CTL	train	73
GSM2630864	CTL	val	74
GSM2630865	CTL	train	75
GSM2630866	CTL	train	76
GSM2630867	PD	train	77
GSM2630868	CTL	train	78
GSM2630869	CTL	train	79
GSM2630870	PD	train	80
GSM2630871	CTL	train	81
GSM2630872	CTL	train	82
GSM2630873	CTL	train	83
GSM2630874	CTL	train	84
GSM2630876	CTL	train	85
GSM2630877	CTL	train	86
GSM2630878	CTL	train	87
GSM2630879	CTL	val	88
GSM2630880	CTL	train	89
GSM2630881	CTL	train	90
GSM2630882	CTL	train	91
GSM2630883	CTL	train	92
GSM2630884	CTL	train	93
GSM2630885	CTL	train	94
GSM2630886	PD	train	95
GSM2630888	CTL	train	96
GSM2630890	PD	train	97
GSM2630891	PD	train	98
GSM2630892	PD	train	99
GSM2630893	PD	train	100
GSM2630895	PD	train	101
GSM2630896	CTL	train	102
GSM2630897	CTL	train	103
GSM2630898	PD	val	104
GSM2630899	CTL	val	105
GSM2630900	PD	val	106
GSM2630901	PD	train	107
GSM2630902	PD	train	108
GSM2630903	PD	val	109
GSM2630904	PD	train	110
GSM2630905	CTL	val	111
GSM2630906	CTL	val	112

continued

GSM	Label	Split	Index
GSM2630907	CTL	val	113
GSM2630908	CTL	train	114
GSM2630909	CTL	val	115
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GSM2630913	CTL	train	119
GSM2630914	PD	train	120
GSM2630915	PD	train	121
GSM2630916	CTL	val	122
GSM2630917	CTL	val	123
GSM2630918	PD	train	124
GSM2630921	PD	train	125
GSM2630922	CTL	val	126
GSM2630923	CTL	val	127
GSM2630924	CTL	val	128
GSM2630925	CTL	val	129
GSM2630927	CTL	val	130
GSM2630928	CTL	val	131
GSM2630929	CTL	val	132
GSM2630930	CTL	val	133
GSM2630931	PD	val	134
GSM2630932	CTL	val	135
GSM2630938	PD	train	136
GSM2630939	PD	train	137
GSM2630940	PD	train	138
GSM2630941	PD	train	139
GSM2630942	PD	train	140
GSM2630944	PD	train	141
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GSM2630950	PD	train	146
GSM2630951	PD	train	147
GSM2630956	PD	val	148
GSM2630957	PD	train	149
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GSM2630959	PD	train	151
GSM2630960	PD	train	152
GSM2630961	CTL	train	153
GSM2630964	PD	train	154
GSM2630965	CTL	train	155
GSM2630966	PD	train	156
GSM2630967	PD	train	157
GSM2630968	PD	train	158
GSM2630969	PD	train	159
GSM2630970	PD	train	160
GSM2630971	CTL	train	161
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GSM2630980	CTL	train	170
GSM2630981	PD	train	171
GSM2630982	PD	val	172
GSM2630983	PD	train	173
GSM2630984	CTL	train	174
GSM2630985	PD	train	175
GSM2630986	PD	train	176
GSM2630988	PD	train	177
GSM2630989	PD	train	178
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GSM2630991	PD	train	180
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GSM2630995	PD	train	184
GSM2630997	PD	train	185
GSM2630998	PD	train	186
GSM2630999	CTL	train	187
GSM2631001	CTL	train	188
GSM2631002	PD	train	189
GSM2631003	PD	train	190
GSM2631004	PD	train	191
GSM2631005	CTL	train	192
GSM2631006	PD	train	193
GSM2631007	PD	train	194
GSM2631008	PD	train	195
GSM2631009	PD	train	196
GSM2631010	CTL	train	197
GSM2631011	CTL	train	198
GSM2631012	CTL	train	199
GSM2631013	PD	train	200
GSM2631014	PD	train	201
GSM2631015	CTL	train	202
GSM2631016	CTL	train	203
GSM2631017	CTL	train	204
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GSM2631025	CTL	train	211
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GSM2631038	CTL	train	223
GSM2631039	CTL	train	224
GSM2631040	CTL	train	225
GSM2631041	CTL	train	226
GSM2631042	CTL	train	227
GSM2631043	CTL	train	228
GSM2631044	CTL	train	229
GSM2631045	CTL	train	230
GSM2631046	CTL	train	231
GSM2631047	CTL	train	232
GSM2631048	PD	train	233
GSM2631049	CTL	train	234
GSM2631050	PD	train	235
GSM2631051	PD	train	236
GSM2631052	PD	train	237
GSM2631053	PD	train	238
GSM2631054	CTL	train	239
GSM2631055	CTL	train	240
GSM2631056	CTL	train	241
GSM2631057	CTL	train	242
GSM2631058	CTL	val	243
GSM2631059	CTL	train	244
GSM2631060	PD	train	245
GSM2631061	CTL	train	246
GSM2631064	CTL	train	247
GSM2631065	CTL	train	248
GSM2631066	PD	val	249
GSM2631067	PD	train	250
GSM2631068	CTL	train	251
GSM2631070	CTL	train	252
GSM2631071	PD	train	253
GSM2631072	CTL	train	254
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GSM2631101	CTL	train	277
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GSM2631103	CTL	train	279
GSM2631104	PD	val	280
GSM2631105	CTL	train	281
GSM2631106	PD	train	282
GSM2631107	PD	train	283
GSM2631108	PD	train	284
GSM2631109	CTL	train	285
GSM2631110	CTL	train	286
GSM2631111	CTL	val	287
GSM2631112	CTL	train	288
GSM2631114	PD	train	289
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GSM2631116	CTL	train	291
GSM2631117	PD	train	292
GSM2631118	CTL	train	293
GSM2631119	PD	train	294
GSM2631120	PD	train	295
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GSM2631122	CTL	train	297
GSM2631123	CTL	train	298
GSM2631124	PD	train	299
GSM2631125	CTL	train	300
GSM2631126	CTL	train	301
GSM2631127	PD	train	302
GSM2631128	PD	train	303
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GSM2631136	CTL	train	311
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GSM2631138	CTL	train	313
GSM2631140	CTL	val	314
GSM2631142	CTL	train	315
GSM2631143	CTL	train	316
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GSM2631153	CTL	val	324
GSM2631154	PD	val	325
GSM2631155	PD	train	326
GSM2631156	PD	train	327
GSM2631157	PD	train	328
GSM2631158	CTL	val	329
GSM2631159	CTL	val	330
GSM2631160	PD	train	331
GSM2631161	CTL	val	332
GSM2631162	CTL	val	333
GSM2631163	CTL	val	334
GSM2631164	PD	val	335
GSM2631165	CTL	val	336
GSM2631166	PD	val	337
GSM2631167	CTL	val	338
GSM2631168	PD	val	339
GSM2631169	PD	train	340
GSM2631170	PD	val	341
GSM2631172	CTL	val	342
GSM2631173	CTL	train	343
GSM2631175	CTL	val	344
GSM2631176	CTL	val	345
GSM2631177	CTL	val	346
GSM2631178	PD	val	347
GSM2631179	CTL	train	348
GSM2631180	CTL	train	349
GSM2631181	CTL	train	350
GSM2631182	PD	train	351
GSM2631183	PD	train	352
GSM2631184	PD	val	353
GSM2631185	CTL	train	354
GSM2631186	PD	train	355
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GSM2631188	PD	val	357
GSM2631189	PD	train	358
GSM2631190	PD	train	359
GSM2631191	CTL	train	360
GSM2631192	PD	train	361
GSM2631194	CTL	train	362
GSM2631195	CTL	train	363
GSM2631196	CTL	train	364
GSM2631197	CTL	train	365
GSM2631198	CTL	train	366
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GSM2631225	PD	test	371
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GSM2631236	CTL	test	378
GSM2631237	CTL	test	379
GSM2631238	CTL	test	380
GSM2631239	CTL	test	381
GSM2631240	CTL	test	382
GSM2631241	CTL	test	383
GSM2631242	CTL	test	384
GSM2631243	CTL	test	385
GSM2631244	PD	test	386
GSM2631245	PD	test	387
GSM2631246	PD	test	388
GSM2631247	PD	test	389
GSM2631248	CTL	test	390
GSM2631249	PD	test	391
GSM2631252	CTL	test	392
GSM2631253	PD	test	393
GSM2631254	PD	test	394
GSM2631256	PD	test	395
GSM2631257	PD	test	396
GSM2631258	PD	test	397
GSM2631259	PD	test	398
GSM2631260	PD	test	399
GSM2631261	PD	test	400
GSM2631262	PD	test	401
GSM2631263	PD	test	402
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GSM2631267	PD	test	406
GSM2631268	PD	test	407
GSM2631269	PD	test	408
GSM2631271	PD	test	409
GSM2631272	CTL	test	410
GSM2631273	CTL	test	411
GSM2631274	CTL	test	412
GSM2631275	CTL	test	413
GSM2631276	CTL	test	414
GSM2631277	CTL	test	415
GSM2631278	CTL	test	416
GSM2631279	CTL	test	417
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GSM2631281	CTL	test	419
GSM2631283	CTL	test	420
GSM2631284	CTL	test	421
GSM2631287	CTL	test	422
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GSM2631293	PD	test	426
GSM2631295	CTL	test	427
GSM2631296	PD	test	428
GSM2631297	PD	test	429
GSM2631298	CTL	test	430
GSM2631300	CTL	test	431
GSM2631301	CTL	test	432
GSM2631304	CTL	test	433
GSM2631305	CTL	test	434
GSM2631306	CTL	test	435
GSM2631310	CTL	test	436
GSM2631312	CTL	test	437